A giant bushfire burning out of control near Cape Town, South Africa in 2009.

UNIT 1

Prosperity, inequality, and planetary limits

How capitalism revolutionized the way we live, and how economics attempts to understand this and other economic systems

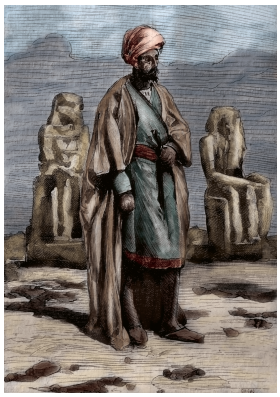
1.1 Ibn Battuta's fourteenth-century travels in a flat world

In the fourteenth century, the Moroccan scholar Ibn Battuta visited the Indian sub-continent. He described the region of Bengal as 'A country of great extent, and one in which rice is extremely abundant. Indeed, I have seen no region of the earth in which provisions are so plentiful.'

And he had seen much of the world, having travelled to China, west Africa, the Middle East, and Europe. Three centuries later, the same sentiment was expressed by the seventeenth-century French diamond merchant, Jean Baptiste Tavernier, who wrote of the country:

> Even in the smallest villages, rice, flour, butter, milk, beans and other vegetables, sugar and sweetmeats, dry and liquid, can be procured in abundance.

At the time of Ibn Battuta's visit to Bengal, Europe was reeling under the impact of the bubonic plague, which took the lives of a quarter or more of the population in many cities. India was not richer than the other parts of the world. But India was not much poorer, either.



Ibn Battuta (1304–1368) travelled across north and west Africa, Europe, the Middle East, south and central Asia and China journeying a total of 117,000 km, almost five times the distance covered by the more famous (also fourteenth-century) traveller Marco Polo.

In our 'Economist in action' video, Thomas Piketty and James Heckman discuss how tax records and other data can be used to provide a social context for our understanding of economics. They argue that this context can create a more detailed narrative for understanding today's economic challenges.

Looking at standards of living around the world, an observer at the time would have noticed that on average people were better off in Italy than in Britain, India, and Japan. But the vast differences between the rich and the poor in any given country were much more striking than these differences across regions. Rich and poor would often have different titles: in some places they would be feudal lords and serfs, in others royalty and their subjects, enslaved people and their owners, or merchants and the sailors who transported their goods. Then—as now—your prospects depended on where your parents were on the economic ladder and whether you were male or female. The difference in the fourteenth century, compared with today, was that back then the part of the world in which you were born mattered much less. Economically speaking the world was flat in Ibn Battuta's day.

Fast forward to today. In many respects, the people of India are better off than they were seven centuries ago—for example in their access to better diets, sanitation, and control of infectious diseases as reflected in greater life expectancy (the number of years a newborn child may expect to live)—but by world standards today, most are poor.

In Ibn Batutta's day, people learned about conditions around the world from the impressions of a few explorers. The data in Figure 1.1, which you will see in the next section, as well as the other figures in this unit, draw on the studies of hundreds of statisticians, historians, computer scientists, and economists. Economics is a fact-based field of study, which uses data to help us understand how the modern world works (and how to make it work better for all).

In our video, the economists Thomas Piketty and James Heckman explain how collecting data has been fundamental to their work on inequality and the policies to reduce it.

1.2 History's hockey stick

Living standards throughout the world have risen dramatically since the time of Ibn Battuta—but much more in some countries than in others.

Figure 1.1 tells only a part of the story. To compare living standards in each country, we start from a measure called **gross domestic product (GDP)**. GDP is a measure of how much is produced in a particular country in a year. We refer to GDP as the 'output' of a country. Diane Coyle, an economist, says it 'adds up everything from nails to toothbrushes, tractors, shoes, haircuts, management consultancy, street cleaning, yoga teaching, plates, bandages, books, and the millions of other services and products in the economy'. These are all added together using their market values, which gives us total output, which also corresponds to the total income of everyone in the country. Then we divide GDP by the total population, and use the resulting number—GDP per capita—to measure average income, or 'living standards'. (Some important things have been left out here. We discuss them in the extension of this section.)

In Figure 1.1, the height of each line is an estimate of average living standards at the date on the horizontal axis. You can see, for example, that in the fifteenth century, living standards were higher in Italy than in any of the other countries for which we have data.

By 2018, according to this measure, people were six times better off, on average, in Japan than in India. People in Japan were nearly as rich as those in Britain, just as they were in the fourteenth century, but people in the US (not shown) were even better off, and people in Norway (also not shown) are better off still.

Listen to Diane Coyle talking about the benefits and limitations of measuring GDP (https://tinyco.re/1216717). You can learn about how GDP is calculated in *The Economy 2.0: Macroeconomics*. In order to compare average living standards across countries and over time, economists use the concept of purchasing power parity (PPP) to account for the differences in prices of goods and services. PPPs are price indices that measure how much it costs to purchase a basket of goods and services compared to how much it costs to purchase the same basket in a reference country in a particular year, such as the United States in 2011.

Before 1300, we have very few data points. For example, we have estimates of Chinese GDP only in 1000, 1090, and 1120, so the graph is drawn by joining these points with straight lines.

We can draw the graph in Figure 1.1 because of the work of Angus Maddison (https://tinyco.re/4376799), who dedicated his working life to finding the scarce data needed to make useful comparisons of how people lived across more than 1,000 years. More recent estimates by economic historians are shown in the figure. This book will show you that the starting point of all economics is data like this about regions of the world, and the people in it.

History's hockey stick does not appear in all countries and, where it does, it is shaped differently for different countries. The hockey stick kink is less abrupt in Britain, where growth began around 1650, while in Japan the kink is sharper, occurring around 1870. In China and India, living standards declined during the period when growth was taking off for countries in western Europe, and the kinks happened much later—in the second half of the twentieth century.

In some economies, including those of China and India, substantial improvements in people's living standards did not occur before they gained independence from colonial rule or interference by European nations.

Figure 1.1 also illustrates that for much of history, living standards did not grow in any sustained way. When sustained growth occurred, it began at different times in different countries, leading to vast differences in living standards between countries around the world. Since late in the twentieth century, 'latecomers' such as India and China have been catching up with the richer nations, but in some countries the hockey stick has not yet tipped upwards.

Understanding why over the past three centuries some countries have prospered and others have not has been one of the most important questions that economists have asked, starting with a founder of the field, Adam Smith. He gave his most important book the title *An Inquiry Into the Nature and Causes of the Wealth of Nations*. Read more in the 'Great economist' feature on Adam Smith.

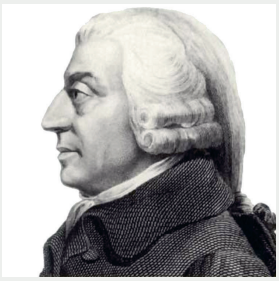
gross domestic product (GDP) A measure of the market value of the output of final goods and services in the economy in a given period. GDP combines in a single number all the output (or production) carried out by the firms, non-profit institutions, and government bodies within a government's territory. Output of intermediate goods that are inputs to final production is excluded to prevent double counting. Household production is part of GDP if it is sold. GDP is measured monthly, quarterly, and annually.

We call these figures 'hockey stick curves' because of their resemblance to the shape of an ice hockey stick.

An entertaining video (https://tinyco.re/3761488) by Hans Rosling, a statistician, shows how some countries got richer—and healthier—much earlier than others.

GREAT ECONOMISTS

Adam Smith



Adam Smith (1723–1790) is considered by many to be the founder of modern economics. Raised by a widowed mother in Scotland, he went on to study philosophy at the University of Glasgow, which was an important centre of Enlightenment thought as well as of colonial trade—both of which influenced his understanding of the economy. He travelled throughout Europe, and while in Toulouse, France, he claimed to have 'very little to do' and so began 'to write a book in order to pass away the time'. This was to become the most famous book in economics.

In *An Inquiry Into the Nature and Causes of the Wealth of Nations*, published in 1776, Smith asked: How can society coordinate the independent activities of large numbers of economic actors—producers, transporters, sellers, consumers—often unknown to each other and widely scattered across the world? His radical claim was that coordination among all of these actors might spontaneously arise, without any person or institution consciously attempting to create or maintain it. This challenged previous notions of political and economic organization, in which rulers imposed order on their subjects.

Even more radical was his idea that this could take place as a result of individuals pursuing their self-interest: 'It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest,' he wrote.

Elsewhere in *The Wealth of Nations*, Smith introduced one of the most enduring metaphors in the history of economics: that of the invisible hand. The businessman, he wrote, 'intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention. Nor is it always the worse for the society that it was no part of it. By pursuing his own interest he frequently promotes that of the society more effectually than when he really intends to promote it.' And he added: 'Nobody but a beggar chooses to depend chiefly upon the benevolence of his fellow-citizens.'

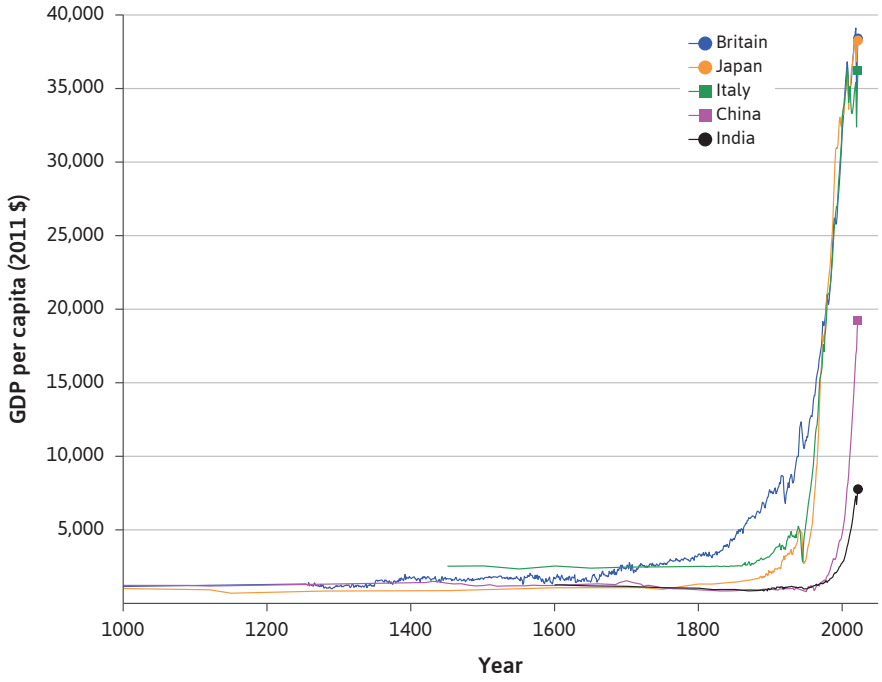


Figure 1.1 History's hockey stick: gross domestic product per capita in five countries (1000–2018).

Explore this data at https://tinyco.re/3290463

Stephen Broadberry. 2021. 'Accounting for the great divergence: recent findings from historical national accounting'.; Total Economy Database (https://tinyco.re/1587851).; S. N. Broadberry, B. Campbell, A. Klein, M. Overton, and B. van Leeuwen, B. 2015. *British Economic Growth, 1270–1870* (https://tinyco.re/6743321). Cambridge: Cambridge University Press.; S. Broadberry, H. Guan, and D. Li. 2018. 'China, Europe and the Great Divergence: A Study in Historical National Accounting' (https://tinyco.re/9902110) *Journal of Economic History* 78: pp. 955–1000.; J. P. Bassino, S. Broadberry, K. Fukao, B. Gupta, and M. Takashima, M. 2019. 'Japan and the Great Divergence, 730–1874' *Explorations in Economic History* 72: pp. 1–22.; S. Broadberry, J. Custodis, and B. Gupta, B. 2015. 'India and the Great Divergence: An Anglo-Indian Comparison of GDP per Capita, 1600–1871' (https://tinyco.re/3221560) *Explorations in Economic History* 55: pp. 58–75.; P. Malanima. 2011. 'The Long Decline of a Leading Economy: GDP in Central and Northern Italy, 1300–1913'. *European Review of Economic History* 15: pp. 169–219.; S. Broadberry and L. Gardner. 2022. 'Economic Growth in Sub-Saharan Africa, 1885–2008: Evidence From Eight Countries' (https://tinyco.re/8802100). *Explorations in Economic History* 83: 101424.
Note: The historical data is being improved continuously and the best data is provided in Figure 1.1 for the six countries shown. An alternative source of data is available for many more countries in the interactive chart.

Among Smith's insights is the idea that a significant source of prosperity is the division of labour, or specialization, and that this in turn is constrained by the 'extent of the market'. Smith illustrated this idea in a famous passage on the pin factory by reporting that ten men, each fully specialized in one or two of 18 distinct operations, could produce close to 50,000 pins a day. But 'if they had all wrought [pins] separately and independently … they certainly could not each of them have made twenty, perhaps not one pin in a day.'

But such an enormous number of pins could only find buyers if they were sold far from their point of production. Hence specialization was fostered by the construction of navigable canals and the expansion of foreign trade. And the resulting prosperity itself expanded the 'extent of the market', in a virtuous cycle of economic expansion.

Smith did not think that people were guided entirely by self-interest. Seventeen years before *The Wealth of Nations*, he had published a book about ethical behaviour called *The Theory of Moral Sentiments*.

He also understood that the market system had some failings, especially if sellers banded together so as to avoid competing with each other. 'People in the same trade seldom meet together,' he wrote, 'even for merriment and diversion, but the conversation ends in a conspiracy against the public; or in some contrivance to raise prices.'

He specifically targeted monopolies that were protected by governments, such as the British East India Company, that not only controlled trade between India and Britain, but also administered much of the British colony there.

He agreed with his contemporaries that a government should protect its nation from external enemies, and ensure justice through the police and the court system. He also advocated government investment in education, and in public works such as bridges, roads, and canals.

QUESTION 1.1 Choose the correct answer(s)

Read the following statements about Adam Smith and choose the correct option(s).

- ☐ Adam Smith believed in the role of the government to improve societal welfare.
- ☐ Adam Smith believed that all markets were characterized by perfect competition.
- ☐ Adam Smith argued that economic agents were guided entirely by self-interest.
- ☐ Adam Smith claimed that coordination among a large number of economic actors (producers, transporters, sellers, consumers), often unknown to one another, might spontaneously arise without any person or institution consciously attempting to create or maintain it.

EXTENSION 1.2

GDP per capita and living standards

In this extension we explore the reasons why GDP is not always a satisfactory measure of living standards. Not only does GDP leave out some things that are important for our daily lives, it also fails to take

Pierre Friedlingstein, Matthew W. Jones, Michael O'Sullivan, et al. 2019. 'Global Carbon Budget 2019' (https://tinyco.re/2898770). *Earth System Science Data* 11: pp. 1783–1838. doi: 10.5194/essd-11-1783-2019.; Pieter Tans NOAA/GML and Ralph Keeling, Scripps Institution of Oceanography. 2022. 'Trends in Atmospheric Carbon Dioxide' (https://tinyco.re/4421890).; D. Gilfillan, G. Marland, T. Boden, and R. Andres, R. 2021. 'Global, Regional, and National Fossil Fuel CO2 Emissions' (https://tinyco.re/3338780). Carbon Dioxide Information Analysis Center (CDIAC) Datasets. Accessed: September, 2021.

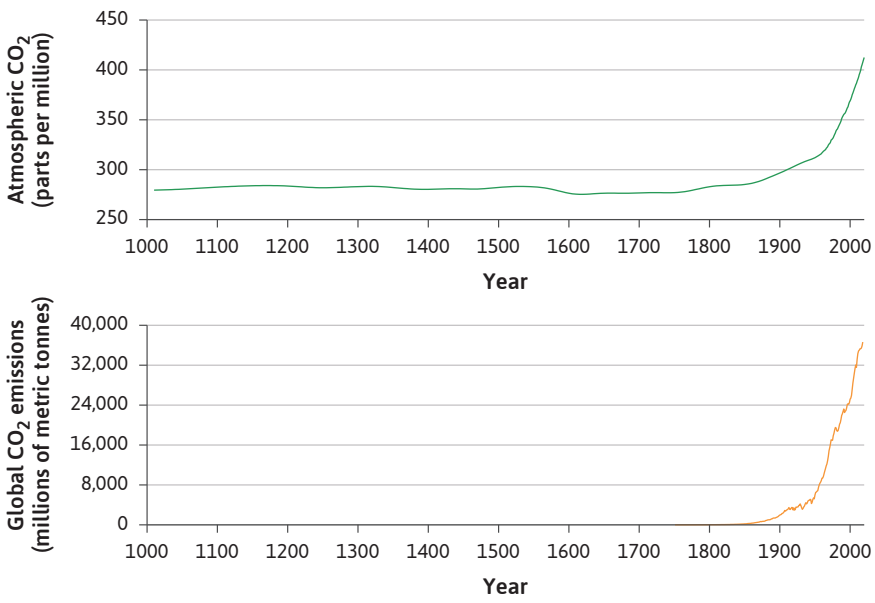


Figure 1.2a Carbon dioxide in the atmosphere (1010–2020) and global carbon emissions from burning fossil fuels (1750–2018).

account of differences between people, and of the depletion of environmental resources.

https://tinyco.re/4703694
Read the whole extension online

1.3 Another hockey stick: Climate change

Using GDP as a measure of living standards ignores the importance of the environment for our current and future wellbeing. Evidence that our use of fossil fuels—coal, oil, and natural gas—has profoundly affected the natural environment of the planet is shown in Figures 1.2a and 1.2b. Each figure has the hockey stick shape. After having remained relatively unchanged for many centuries, increasing emissions of carbon dioxide (CO_2) into the air during the twentieth and twenty-first centuries have resulted in measurably larger amounts of CO_2 in the earth's atmosphere (Figure 1.2a) and brought about perceptible increases in the northern hemisphere's average temperatures (Figure 1.2b). Figure 1.2a also shows that CO_2 emissions from fossil fuel consumption have risen dramatically since the late 1800s.

Figure 1.2b shows that the mean temperature of the earth fluctuates from decade to decade. Many factors cause these fluctuations, including volcanic events such as the 1815 Mount Tambora eruption in Indonesia.

Since 1900, average temperatures have risen in response to increasingly high levels of greenhouse gas concentrations. These have mostly resulted from the CO_2 emissions associated with the burning of fossil fuels. And in each year of the twenty-first century, the average temperature has been higher than at any time in the previous millennium.

Mount Tambora spewed so much ash that the earth's temperature was reduced by the cooling effect of these fine particles in the atmosphere, and 1816 became known as the 'year without a summer' (https://tinyco.re/3530430).

The authoritative source for research and data about climate change is the Intergovernmental Panel on Climate Change (https://tinyco.re/8844088).

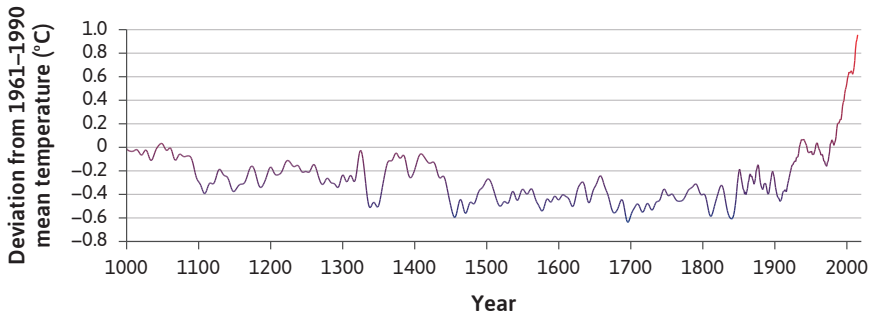


Figure 1.2b Northern hemisphere temperatures over the long run (1000–2019). The figure shows 5-year moving averages.

Explore this data at https://tinyco.re/8926412

Michael E. Mann, Zhihua Zhang, Malcolm K. Hughes, Raymond S. Bradley, Sonya K. Miller, Scott Rutherford, and Fenbiao Ni. 2008. 'Proxy-based reconstructions of hemispheric and global surface temperature variations over the past two millennia' (https://tinyco.re/1009800). Proceedings of the National Academy of Sciences 105 (36): pp. 13252–13257.; C. P. Morice, J. J. Kennedy, N. A. Rayner, and P. D. Jones. 2012. 'Quantifying uncertainties in global and regional temperature change using an ensemble of observational estimates: The HadCRUT4 dataset' (https://tinyco.re/6765840), Journal of Geophysical Research 117. D08101, doi:10.1029/2011JD017187.

The human causes and the reality of climate change are no longer widely disputed in the scientific community. The likely consequences of global warming are far-reaching: melting of the polar ice caps, rising sea levels that may put large coastal areas under water, and potential changes in climate and rain patterns that may make some densely populated parts of the world uninhabitable and destroy the world's food-growing areas.

We can see that the hockey sticks for GDP per capita and for atmospheric CO_2 have risen together. It is also the case that richer countries have, on average, higher emissions per capita. In Unit 2 (page 0), we explore this link between income and emissions, and consider whether it will be possible, in future, to raise living standards around the world without further damage to the climate.

EXERCISE 1.1 How much difference does a couple of degrees warmer or colder make?

Between 1300 and 1850 there were a number of exceptionally cold periods, as shown in Figure 1.2b. Research this so-called 'little ice age' in Europe and answer the following questions.

1. Describe the effects of these exceptionally cold periods on the economies of these countries.
2. Provide examples of groups of people within a country or region who were exceptionally affected by climate change.

Michail Moatsos. 2021. 'Global extreme poverty: Present and past since 1820' (https://tinyco.re/6574838). Published in OECD. 2021. *How Was Life? Volume II: New Perspectives on Well-Being and Global Inequality Since 1820* (https://tinyco.re/8763321), OECD Publishing, Paris.

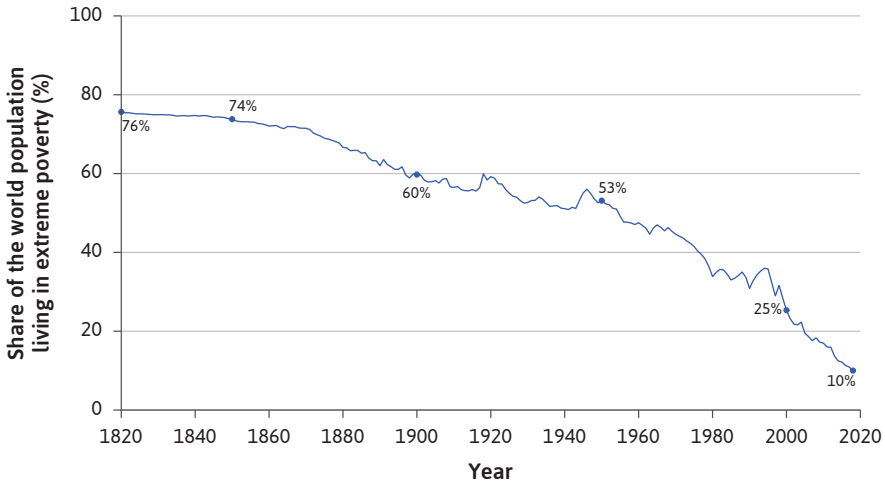


Figure 1.3 Share of the world population living in extreme poverty, 1820–2018.

QUESTION 1.2 Choose the correct answer(s)

Figure 1.2b shows the northern hemisphere's temperature since year 1000, reported as the deviation from the 1961–1990 mean (average) temperature. Based on this figure, read the following statements and choose the correct option(s).

- ☐ The 1961–1990 mean temperature was 0.2 to 0.6 degrees higher than the temperatures between 1450 and 1900.
- ☐ The negative numbers on the graph indicate that the temperature consistently fell between 1100 and 1900.
- ☐ A consistent rise in temperature is only a post-1980 phenomenon.
- ☐ The consistent rise in temperature after 1980 suggests that temperatures will continue to rise in every year following 2000.

1.4 Inequality in global income

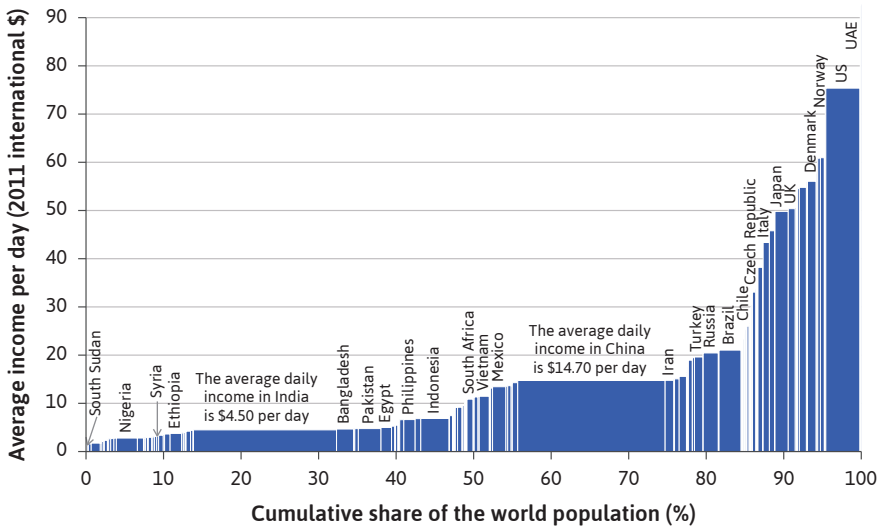
Before the upward kink in history's hockey stick, some people were rich by the standards of their day, but most people in the world lived in extreme poverty.

Global poverty and income inequality

To explore the latest data on extreme poverty, read Our World in Data's webpage on 'Poverty' (https://tinyco.re/9647535).

Mirroring the hockey sticks of income and carbon is the transformation from a world in which the vast majority of people were unable to meet their basic needs of minimal nutrition and adequately heated shelter to today, when that is true of around one in ten (Figure 1.3). These people are often hungry, don't have access to electricity (and therefore to light at night, for example), and have limited access to schooling or healthcare.

A thousand years ago, the world was flat, economically speaking. There were major differences in income within the world's countries and regions; but as shown in Figure 1.1 (page 4), the differences between countries were small, especially when compared to what was to follow.



M. Roser. 2021. 'Global poverty in an unequal world: Who is considered poor in a rich country? And what does this mean for our understanding of global poverty?' (https://tinyco.re/7735461) Our World in Data. 'Income inequality within and between countries'.; The World Bank. 2022. Poverty and Inequality Platform. 'Mean income or expenditure per day' (https://tinyco.re/2334110).; The World Bank. 2022. 'Population, total' (https://tinyco.re/1498000).
Note: Average income is measured in 'international dollars', which enables comparisons of the purchasing power of incomes to be made across countries; this is known as purchasing power parity.

Figure 1.4 The global income distribution in 2019: average daily income by country.

Income inequality between countries

Nobody thinks the world is flat today, when it comes to income. Figure 1.4 provides a snapshot for 2019. The countries of the world are lined up from left to right by average income, the poorest (South Sudan) on the left and the richest (United Arab Emirates) on the right. The width of the bars reflects the size of the population, with the bars for China and India being the widest. The horizontal scale shows the cumulative share of the world's population, with 100% at the far right. On the vertical axis is the average daily income—measured in such a way that $20, for example, is what a person in any given country can buy in goods and services that cost $20 in the US. Economists measure prices using **purchasing power parity** to make these comparisons. In the bottom 10% of countries, where people on average live in extreme poverty, the daily income is just a few dollars a day. The snapshot provided by average incomes shows big differences between countries, but it does not show the vast differences in living standards within each country.

Figure 1.1 (page 4) showed the great divergence in fortunes across the world beginning around the seventeenth century and taking off in the eighteenth. Extreme poverty has shrunk. But stark inequalities in income persist within countries—and have increased in recent decades. While Figure 1.4 shows the average income in each country, Figure 1.5 goes a step further by building a 3D picture that shows the distribution of income *within countries* as well as between them, and how this changed between 1980 and 2020.

Again, countries are arranged from the poorest (according to average income in that year) on the left of the diagram (South Sudan in 1980) to the richest on the right (Switzerland in 1980), with the width of each country's bars representing its population.

For every country there are ten bars, corresponding to the ten **decile groups** of income. Each group contains 10% of the population, and the height of the bar is the average income of the group, ranging from the poorest 10% of people at the front of

For example, 14% of the world's population live in countries with lower average income than India. Since 18% of the world's population live in India, a total of 32% live in countries with average income less than or equal to that in India.

purchasing power parity (PPP) PPPs are price indices that measure how much it costs to purchase a basket of goods and services compared to how much it costs to purchase the same basket in a reference country in a particular year, such as the United States in 2011.

decile, decile groups
Deciles split a set of observations into ten equally-sized groups. The full set of observations is ordered according to a particular variable (e.g. income). The first decile group is the observations in the bottom 10% (e.g. the 10% with the lowest incomes), the second is the next lowest 10%, and the tenth or top decile group is the highest 10%. The deciles are the cutoff values that separate the groups; the first decile is the cutoff between the first and second decile groups, and so on.

the diagram to the richest 10% at the back, measured in 2021 US dollars. Note that this measure doesn't mean 'the richest 10% of income earners'. It is the richest 10% of people, where each person in a household, including children, is assumed to have an equal share of the household's income.

The skyscrapers (the highest columns) at the back of the right-hand side of the figure represent the income of the richest 10% in the richest countries. In 2020, the tallest skyscraper represents the richest 10% of people in the United Arab Emirates. In 2020, this exclusive group had an income per capita of around $390,000. Norway, the country with the second-highest GDP per capita, does not have a particularly tall skyscraper (it is hidden between the skyscrapers for the United Arab Emirates and the third-richest country, the US) because income is more evenly distributed in Norway than in some other rich countries. Go through the steps in Figure 1.6 to understand how income within and between countries has changed over time.

Two things are clear from the 2020 distribution. First, in every country, the rich have much more than the poor. We can use the ratio between the heights of the front and back bars as one measure of inequality in a country. We will call it the rich/poor ratio, for obvious reasons. Even in a relatively equal country such as Norway, the rich/poor ratio is 66; in the US, it is 244; and in Botswana in southern Africa, it is 489. Inequality within the very poorest countries is difficult to determine from the graph because the height of all the bars are much smaller than those of the richer countries, but inequality is definitely there: the rich/poor ratio is 174 in Nigeria, and 246 in India.

The second thing that jumps out from Figure 1.5 is the huge difference in income between countries. In 2020, the average income in Norway is nearly seven times the average income in Nigeria.

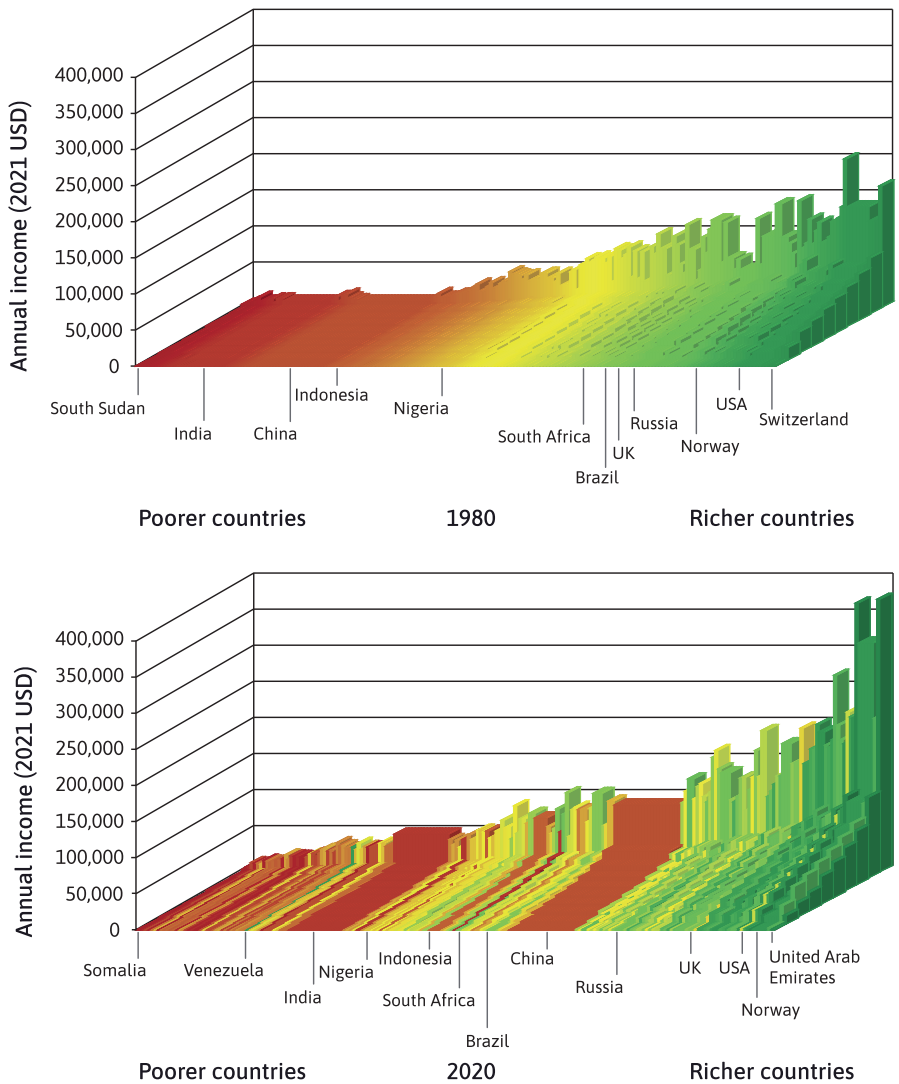
Imagine the traveller Ibn Battuta's journey across regions of the world in the fourteenth century and think of how a diagram like Figure 1.5 (page 11) would have looked then. He would, of course, notice that everywhere he went there were differences between the richest and poorest groups in the population of each region. He would report back that the differences in income between the countries of the world were relatively minor by comparison.

The vast differences in income between the countries of the world today take us back to Figure 1.1 (page 4), where we can begin to understand how this came about. The countries that took off economically before 1900, such as Britain, Japan, and Italy, are now rich. They (and countries like them) are in the skyscraper part of Figure 1.5. The countries that took off only recently, or not at all, are in the flatlands.

What lies behind the upward kink in the hockey sticks that we have seen in some countries but not others, and the accompanying changes in poverty and inequality? We begin to answer this question in the next section.

EXERCISE 1.2 Inequality in the fourteenth century
Ibn Battuta (1304–1368), who we discussed in Section 1.1, was a Moroccan traveller and merchant. His travels, which lasted 30 years, took him across north and west Africa, eastern Europe, the Middle East, south and central Asia, and China.

What do you think a 'skyscraper' figure like that in Figure 1.5 would have looked like at the time of Ibn Battuta (early to mid-fourteenth century)? (Use his descriptions of the countries he visited (https://tinyco.re/3808057) to support your answer.)



GCIP. 2015. Global Consumption and Income Project (https://tinyco.re/8468009). Bob Sutcliffe designed the representation of global inequality in Figure 1.5. A first version was published in: Robert B. Sutcliffe. 2001. *100 Ways of Seeing an Unequal World*. London: Zed Books. An interactive visualization (https://tinyco.re/4877569) of it is available.

Figure 1.5 Global income inequality 1980 (top panel) and 2020 (bottom panel). Countries are ranked by income per capita from left to right. For each country, the heights of the bars show average income for deciles groups of the population, from the poorest 10% at the front to the richest 10% at the back. The width of the bar indicates the country's population.

GCIP. 2015. Global Consumption and Income Project (https://tinyco.re/8468009). Bob Sutcliffe designed the representation of global inequality in Figure 1.6. A first version was published in: Robert B. Sutcliffe. 2001. *100 Ways of Seeing an Unequal World.* London: Zed Books. An interactive visualization (https://tinyco.re/4877569) of it is available.

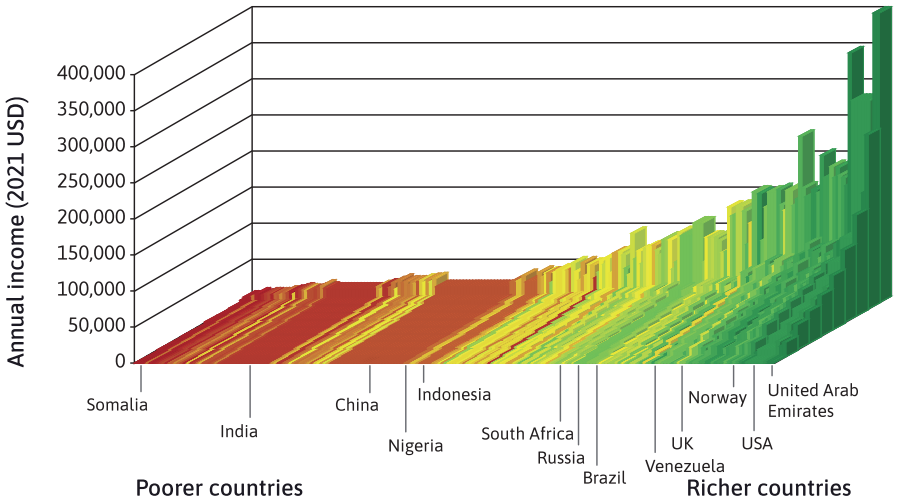


Figure 1.6 Understanding the global income distribution.

1. The richest and poorest countries in 2020

In the United Arab Emirates, the richest country, on the furthest right, the average incomes of the richest and poorest 10% are $389,750 and $1,522 respectively. In Somalia, the furthest left, the corresponding incomes are $6,894 and $16.

2. Skyscrapers

The skyscraper bars in the back right of the figure are the richest 10% in some of the richest countries.

3. World income distribution in 1980 (Figure 1.5, top panel)

In 1980 the ranking of countries by GDP was different. The poorest countries, coloured darkest red, were South Sudan, Somalia, and Mozambique. The richest (darkest green) were Switzerland, Netherlands, and Germany. At that time the skyscrapers were not as tall: the differences between the richest 10% and the rest of a country's population were not as pronounced.

4. World income distribution in 2000

The colours indicate how some countries changed their ranking between 1980 and 2000. Both China and Indonesia used to be relatively poor (red) but have now moved towards the middle of the distribution. Some taller skyscrapers have appeared at the back: inequality increased in many countries during the 1980s and 1990s.

5. World income distribution in 2020 (Figure 1.5, bottom panel)

By 2020, many countries have changed their ranking. China has grown rapidly since 2000. But the countries that were richest in 1980 (darkest green), such as the US and Norway, are still near the top in 2020.

6. Inequality within countries has risen

Income distributions have become more unequal in many of the richer countries: some very tall skyscrapers have appeared. In the middle-income countries, too, there is a big step up at the back of the figure: the incomes of the richest 10% are now high relative to the rest of the population.

QUESTION 1.3 Choose the correct answer(s)

Figure 1.5 shows the global income distribution in 1980 and 2020. Based on this figure, read the following statements and choose the correct option(s).

- ☐ China and Indonesia had similar average incomes in 1980.
- ☐ From 1980 to 2020, income distributions have become more unequal in many countries.
- ☐ The rankings of countries by average income have remained similar from 1980 to 2020.
- ☐ Indonesia and Brazil had a similar population size in 2020.

EXERCISE 1.3 Working with income data

Go to our interactive visualization (https://tinyco.re/4877569) which contains the data used to create Figure 1.5. Choose five countries that you are interested in.

1. In the visualization, use the 'Rich/poor income ratios' tab to find the rich/poor ratio in 1980, 2000, and 2020 for each of your chosen countries.
2. Describe the differences between countries and the changes over time that you find.
3. Suggest some explanations for your answers to question 2.

1.5 The continuous technological revolution

The science fiction show *Star Trek* is set in the future, when humans travel the galaxy with friendly aliens, aided by intelligent computers, faster-than-light propulsion, and replicators that create food and medicine on demand. Whether we find the stories silly or inspiring, most of us, in optimistic moods, can entertain the idea that the future will be transformed morally, socially, and materially by technological progress.

No *Star Trek* future awaited the grandchildren of peasants in 1250. The next 500 years would pass without any measurable change in the standard of living for ordinary working people. While science fiction began to appear in the seventeenth century (Francis Bacon's *New Atlantis* being one of the first works of science fiction, in 1627), it was not until the eighteenth century that each new generation could look forward to a different life, shaped by new technology.

A remarkable number of scientific and technological advances coincided with the upward kink in the income hockey stick in Britain in the middle of the eighteenth century.

Important new technologies were introduced in textiles, energy, and transportation. As late as 1800, traditional craft-based techniques, using skills that had been handed down from one generation to the next, were still used in most production processes. The new era brought new ideas, new discoveries, new methods, and new machines—and made old ideas and old tools obsolete. These new ways were, in turn, made obsolete by even newer ones.

This period of cumulative innovation is called the **Industrial Revolution**.

In everyday usage, 'technology' refers to machinery, equipment, and devices developed using scientific knowledge. In economics, a **technology** is a process that takes a set of materials and other inputs—including the work of people and machines—and creates an output. For example, a technology for making a cake can be described by the recipe specifying the combination of inputs (ingredients such as

Industrial Revolution A wave of technological advances and organizational changes that began in Britain in the eighteenth century; it transformed an agricultural and craft-based economy into a commercial and industrial economy.

technology The description of a process that uses a set of materials and other inputs, including the work of people and machines, to produce an output.

flour, and activities like stirring) needed to create the output (the cake). Another cake-making technology uses large-scale machinery, ingredients, and labour (machine operators).

technological progress A change in technology that reduces the amount of resources (labour, machines, land, energy, time) required to produce a given amount of the output.

Until the Industrial Revolution, the economy's technology, like the skills needed to follow its recipes, was updated only slowly and passed from generation to generation. As **technological progress** revolutionized production, the time required to make a pair of shoes halved within a few decades; the pattern was similar for spinning and weaving, and making cakes in a factory. This marked the beginning of a continuous technological revolution: the time required to produce most products fell from one generation to the next.

For most of human history, technological progress in lighting was slow. Our distant ancestors had nothing brighter than a campfire at night. Thousands of years later, in the early 1800s, tallow candles provided about nine times as much light for an hour of labour as the animal fat lamps of the past. New technologies since then have rapidly increased lighting efficiency. Three decades ago in 1992, compact fluorescent bulbs were introduced and were about 45,000 times more efficient than lights were 200 years ago.

The continuous technological revolution

David Landes, an economic historian, wrote that the Industrial Revolution was 'an interrelated succession of technological changes' that transformed the societies in which these changes took place. Figure 1.7 illustrates some of these changes during the Industrial Revolution in Britain between 1700 and 1850, and the subsequent period of rapid technological developments in other countries, particularly the US, with the accompanying changes in living standards.

An early invention was Thomas Newcomen's steam engine, which was used to pump water out of mines. Developed further by James Watt, steam engines were eventually used across the economy: in textiles, manufacturing, railways, and steamships. They are an example of what is termed a general-purpose innovation or

Jutta Bolt and Jan Luiten van Zanden. 2020. Maddison Project Database, version 2020. 'Maddison style estimates of the evolution of the world economy. A new 2020 update' (https://tinyco.re/7763210)

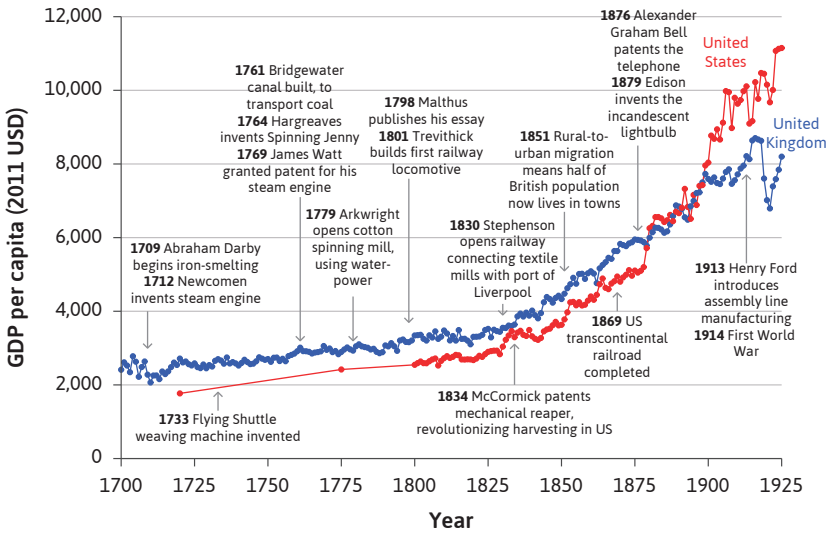


Figure 1.7 The Industrial Revolution and the continuous technological revolution.

technology: a technological advancement that can be applied to many sectors, and which spawns further innovations.

The process of innovation did not end with the Industrial Revolution, as the example of lighting shows. It has continued with the application of new technologies in many industries, including electricity, transportation (canals, railroads, automobiles), and most recently, the revolution in information processing and digital communication. These broadly applicable general-purpose innovations give a particularly strong impetus to growth in living standards, because they change the way that large parts of the economy work.

The number of internet users worldwide rose from 2.6 million in 1990 to 4.7 billion by 2020.

But if technological progress raised living standards as we have suggested, then why was the hockey stick flat for most of the past thousand years? This unit and the next will show how economic reasoning helps to explain both the flat part of the hockey stick and the upturn.

QUESTION 1.4 Choose the correct answer(s)

Which of the following variable(s) have followed the so-called 'hockey stick' trajectory —that is, little to no growth for most of history followed by a sudden and sharp change to a positive growth rate?

- ☐ GDP per capita
- ☐ technological progress in lighting
- ☐ income inequality
- ☐ atmospheric CO_2

1.6 Explaining the flat part of the hockey stick: Production functions and the diminishing average product of labour

Before 1800, most people in most countries made their living in agriculture. In 1798 an English clergyman, Thomas Robert Malthus, published *An Essay on the Principle of Population*, which presented a pessimistic view of the potential for technological progress to raise living standards.

In Italy, for example, the proportion of the labour force in agriculture stayed almost constant at around 60% between 1300 and 1800. In 2019, it was less than 4%.

His argument was based on a simple model of the economy, in which agricultural output and living standards depend on the population employed in agriculture, and in turn, those living standards affect the growth of the population. Malthus argued that a sustained increase in income per capita would be impossible. We now know that he was wrong, but his analysis still helps us to understand why living standards had remained low for hundreds of years before his time.

A model is a simplified way of understanding the economy. We explain more about models in Unit 2 (page 0).

In this section, we explain his model of the relationship between employment and output, which introduces concepts that are used widely in economics—in particular the production function and the diminishing average product of labour.

factor of production Any input into a production process is called a factor of production. Factors of production may include labour, machinery and equipment (usually referred to as capital), land, energy, and raw materials.

A model of production

Imagine an agricultural economy that produces just one good: grain. Suppose that grain production is very simple—involving only farm labour to work the land. Ignore the need for spades, combine harvesters, grain elevators, silos, and other types of buildings and equipment; labour and land are the **factors of production** in this model, and therefore the inputs into the production process.

But the quantity of land is fixed. So if employment is increased, there will be more workers on the same quantity of land, and the average output of grain per worker will fall. In other words, the average product of labour diminishes.

average product The average product of an input is total output divided by the total amount of the input. For example, the average product of a worker (also known as labour productivity) is total output divided by the number of workers employed to produce it.

To be more precise, suppose there are 800 farmers, and the land is divided equally between them. All land is of the same quality. Each farmer works the same total number of hours during a year. Together, the 800 farmers produce a total of 504,000 kg of grain. The **average product** of a farmer's labour is:

$$\text{average product of labour} = \frac{\text{total output}}{\text{total number of farmers}}$$
$$= \frac{504{,}000 \text{ kg}}{800 \text{ farmers}}$$
$$= 630 \text{ kg per farmer}$$

How does grain production in this economy change if the number of farmers changes? Figure 1.8a lists some values of the labour input and corresponding grain output. In the third column, we have calculated the average product of labour.

PRODUCTION FUNCTION
A production function is a graphical or mathematical description of the relationship between the quantities of the inputs to a production process and the amount of output produced.

We call the relationship between the inputs to a production process and the amount of output a **production function**. In this example there are two inputs: labour and land. But since the amount of land is fixed, we can represent the production function as in Figure 1.8b–by plotting a graph that shows the number of farmers on the horizontal axis, and the corresponding output of grain on the vertical axis (assuming that the relationship holds for the numbers in between those shown in the table).

Labour input (number of farmers)	Grain output (kg)	Average product of labour (kg/farmer)
200	171,000	855
400	300,000	750
600	409,000	682
800	504,000	630
1,000	587,000	587
1,200	659,000	549
1,400	723,000	516
1,600	778,000	486
1,800	825,000	458
2,000	864,000	432
2,200	895,000	407
2,400	919,000	383
2,600	935,000	360
2,800	944,000	337
3,000	946,000	315

Figure 1.8a The relationship between the input of labour and the output of grain.

You can think of the production function as an 'if–then' statement: if there are *X* farmers, then they will harvest *Y* kg of grain. Expressing it mathematically, we say that '*Y* is a function of *X*' :

$$Y = f(X)$$

X is the amount of labour devoted to farming, *Y* is the output in grain that results from this input, and the function $f(X)$ describes the relationship between *X* and *Y*, represented by the curve in Figure 1.8b. Work through the steps in the figure to see how to interpret the graph.

The calculations in the third column of Figure 1.8a demonstrate that the more farmers there are, the lower is the average product of labour. Figure 1.8c shows that the average product of labour at each point on the graph corresponds to the slope of the line from the point to the origin. For example, the slope at point B is the vertical distance between the origin and B (778,000) divided by the horizontal distance (1,600), and the average product when there are 1,600 farmers is 778,000/1,600 = 486.

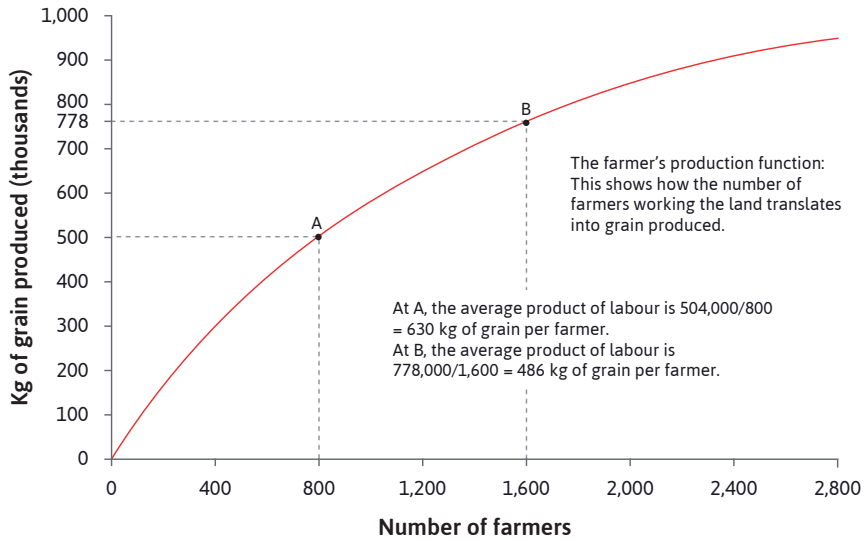


Figure 1.8b The farmers' production function.

1. The farmers' production function
The production function shows how the number of farmers working the land translates into grain produced at the end of the growing season.

2. Output when there are 800 farmers
Point A on the production function shows the output of grain produced by 800 farmers.

3. Output when there are 1,600 farmers
Point B on the production function shows the amount of grain produced by 1,600 farmers.

4. The average product
At A, the average product of labour is 504,000/800 = 630 kg of grain per farmer.
At B, the average product of labour is 778,000/1,600 = 486 kg of grain per farmer.

The line from A to the origin is steeper (the slope is higher). So without doing any calculations, we can tell from the graph that the average product is higher at A. Also, if you draw the lines to the origin from other points they will get flatter as you move along the graph—that is, as the number of farmers increases. This happens because of the concave shape of the production function: it continues to rise but gets flatter as more farmers work the land.

Our grain production function is hypothetical, but it has two features that are plausible assumptions about farming:

- Labour combined with land is productive. The more farmers there are, the more grain is produced (up to 3,000 farmers, at least).
- As more farmers work on a fixed quantity of land, the average product of labour falls. This diminishing average product of labour is one of the foundations of Malthus's model (and many other models).

The diminishing average product of labour worried Malthus.

To understand why, imagine that a generation later, each of the 800 farmers has had many children. So instead of a single farmer working each farm, there are now two farmers. The total labour input into farming was 800, but is now 1,600. The average grain produced per farmer has fallen from 630 kg to 486 kg.

You might argue that in reality, as the population grows, more land can be used for farming. But Malthus pointed out that earlier generations of farmers would have picked the best land, so any new land would not be as productive. This also reduces the average product of labour.

A diminishing average product of labour in agriculture can be caused by:

- more labour devoted to a fixed quantity of land
- more (inferior) land brought into cultivation

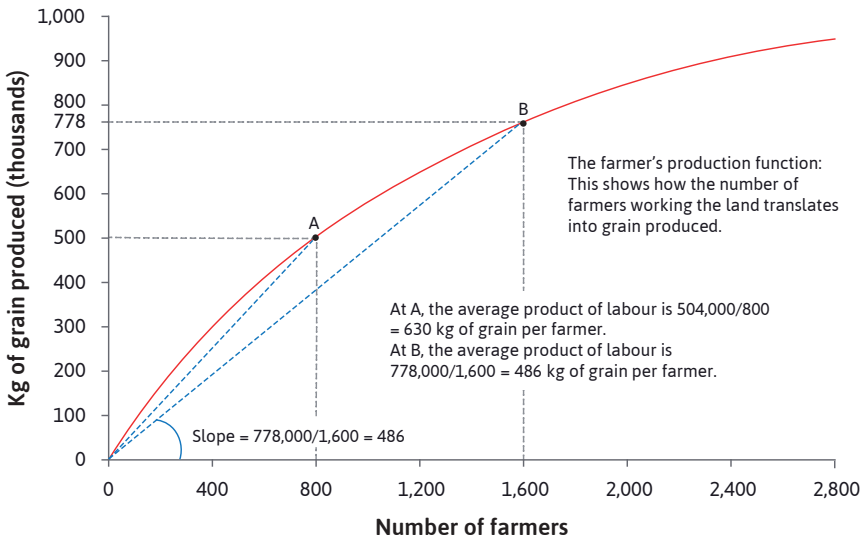


Figure 1.8c The diminishing average product of labour.

Because the average product diminishes as more labour is devoted to farming, the amount of grain each person can consume (their income or living standard) inevitably falls.

EXERCISE 1.4 The farmer's production function
Think of farming biologically.

1. Find out how many calories per hour a farmer burns on average, and how many calories are contained in 1 kg of grain.
2. Does farming produce a surplus of calories—more calories in the output than used up in the work input—at points A and B on the production function in Figure 1.8b? Clearly state any assumptions you make in your calculations.

QUESTION 1.5 Choose the correct answer(s)
Figure 1.8b depicts the production function of grain for farmers under average growing conditions with the currently available technology. Based on this information, we can ascertain that:

- ☐ In a year with exceptionally good weather conditions, the production function curve will be higher and parallel to the curve above.
- ☐ A discovery of new high-yielding crop seeds would tilt the production function curve higher, pivoted anticlockwise at the origin.
- ☐ In a year of bad drought, the production curve can slope downwards for large numbers of farmers.
- ☐ If there is an upper limit on the total amount of grain that can be produced in the country, then the curve will end up horizontal for large numbers of farmers.

1.7 Explaining the flat part of the hockey stick: The Malthusian trap, population, and the average product of labour

On its own, the diminishing average product of labour cannot explain the long, flat portion of the hockey stick. It just means that living standards depend on the size of the population. It doesn't tell us why, over long periods, living standards and population didn't change much. But Malthus's model contains a second key idea:

- Population expands if living standards increase.

His logic was that, if agricultural technology improved and raised the productivity of labour, people would have more children as soon as they were somewhat better off. Because the quantity of land and other available natural resources would not expand along with population, the result would be to lower how much each person could produce. But as long as people were better off, population growth would continue until living standards fell to a sufficiently low level to halt the population increase. Among economists, Malthus's idea of a poverty trap, illustrated in Figure 1.9, was widely accepted.

It provided an explanation of the world in which they lived, in which incomes might fluctuate from year to year (or even century to century), but not permanently trend upwards. This had been the case in many countries for at least 700 years before

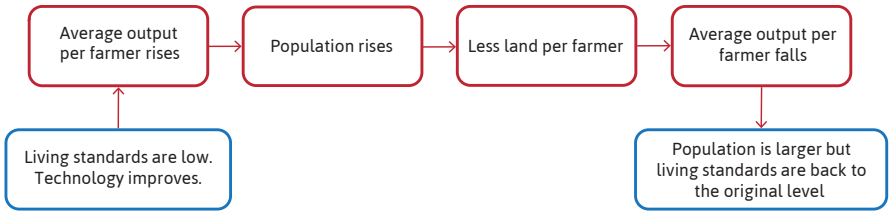


Figure 1.9 Malthus's argument: why technological improvement in farming doesn't raise living standards.

Malthus published his essay, as shown in the long flat part of the hockey stick in Figure 1.1 (page 4).

His theory of population essentially regarded people as not that different from other animals. He was not the first person to have this idea. Years earlier, the Irish economist, Richard Cantillon had stated that, 'Men multiply like mice in a barn if they have unlimited means of subsistence.'

Imagine a herd of antelopes on an otherwise empty plain (there are no predators). When the antelopes are better fed, they live longer and have more offspring. When the herd is small, the antelopes can eat all they want, and the herd gets larger.

Eventually the herd will get so large (relative to the size of the plain), that the antelopes can no longer eat all they want. As the quantity of land per animal declines, they have less to eat. This reduction in living standards will continue as long as the herd continues to increase in size.

Since each antelope has less food, they will have fewer offspring and die younger. Population growth will slow down. Eventually, living standards will be so low that the herd no longer increases in size. The antelopes have filled up the plain. At this point, each animal will be eating an amount of food that we call the **subsistence level**.

The Malthusian population model might appear at odds with what we observe in many higher-income countries today: those with higher income tend to have fewer, not more children. But to explain the flat part of the hockey stick, we need to think about a time when people were much poorer and most people desired larger families than they could afford to raise, so when income went up, the size of a family grew.

subsistence level The level of living standards (measured by consumption or income) below which the population will decline.

Much of the same logic would apply, Malthus reasoned, to a human population with a fixed supply of agricultural land. While people were well fed, they would multiply like Cantillon's mice; but eventually they would fill the country, and further population growth would push down the incomes of most people as a result of diminishing average product of labour. Falling living standards would slow population growth as death rates increased and birth rates fell; ultimately incomes would settle at the subsistence level.

In Figure 1.10, we have plotted the relationship between the average product of labour and the number of farmers in our model of an agricultural economy, using the table in Figure 1.8a (page 16). When the number of farmers is low, their average product is high: for example, if 200 farmers work the land, the output is 855 kg of grain per farmer. As the number of farmers increases, the average product falls. So the graph of the average product of labour is a downward-sloping line.

Suppose that the subsistence level of income in this economy is 500 kg of grain per farmer. In Figure 1.10, the average product is equal to subsistence income when there are 1,500 farmers, at point E. At this point the population neither grows nor falls: it remains constant. If the number of farmers is below 1,500, their income is above subsistence level—they have a higher standard of living, but then the population will rise. And as it rises, the number of farmers working the land will

increase and their average product will fall. So eventually the economy will reach point E, with 1,500 farmers each receiving subsistence income. And if the number of farmers is above 1,500, the opposite will happen: since incomes are below subsistence level, the population will fall, until there are 1,500 farmers and incomes reach subsistence level.

Figure 1.10 captures the essence of Malthus's argument: the economy will tend to return to an **equilibrium** level of population and income, at which income is equal to subsistence level. Equilibrium is an important concept in economics. In everyday language, something is in equilibrium if the forces acting on it are in balance, so it remains unchanged. The same is true here: point E is an equilibrium because if income and population take the values at point E, there is no tendency for them to change; they will remain constant at these values (unless something else in the economy changes).

equilibrium An equilibrium is a situation or model outcome that is self-perpetuating: if the outcome is reached it does not change, unless an external force disturbs it. By an 'external force', we mean something that is determined outside the model.

EXERCISE 1.5 Are people really like other animals?

Malthus wrote: '[I]t is not to be supposed that the physical laws to which [humankind] is subjected should be essentially different from those which are observed to prevail in other parts of the animated nature.'

Are there any differences between humans and animals that could be relevant to Malthus's argument? (It may be helpful to think about the analogy of antelopes on a plain that was used in this section to explain the relationship between population growth and living standards—in which ways does the analogy not apply to humans in the same way as it does to animals?)

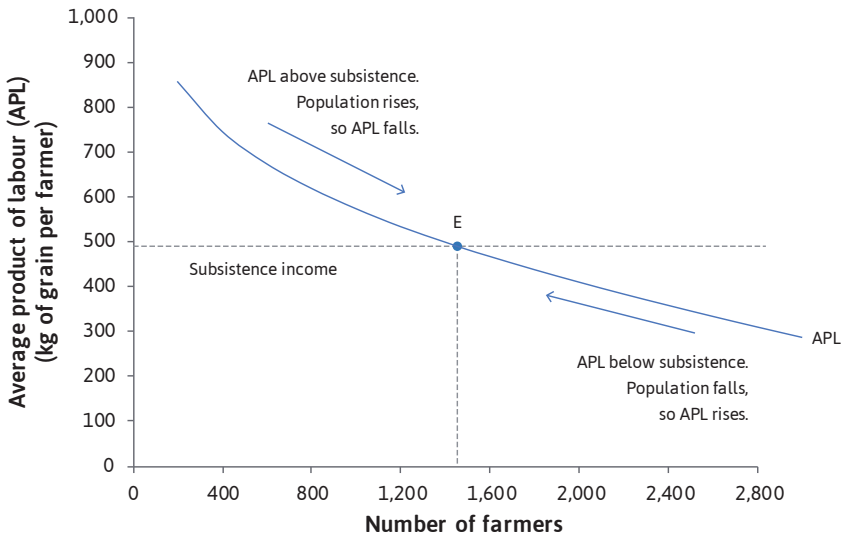


Figure 1.10 Equilibrium in a Malthusian model of an agricultural economy.

Malthusian economics: The effect of technological improvement

We know that over the centuries before the Industrial Revolution, improvements in technology occurred in many regions of the world, including Britain, and yet living standards remained constant. Can Malthus's model explain this?

Figure 1.11 shows what happens in our model following an improvement in agricultural technology—such as the new type of plough from China that came to England in the early seventeenth century. The average product of labour rises at each level of the labour input. In the figure, the line representing the average product of labour shifts upwards.

Suppose that the economy is initially in equilibrium at point E, with 1,500 farmers and subsistence income of 500 kg of grain per farmer. Follow the steps in Figure 1.11 to understand what happens when productivity rises.

Technological progress has raised living standards only temporarily. At the new equilibrium, the population is higher—but income has returned to subsistence level. Rather than raising living standards, a better technology provides subsistence income for a larger population.

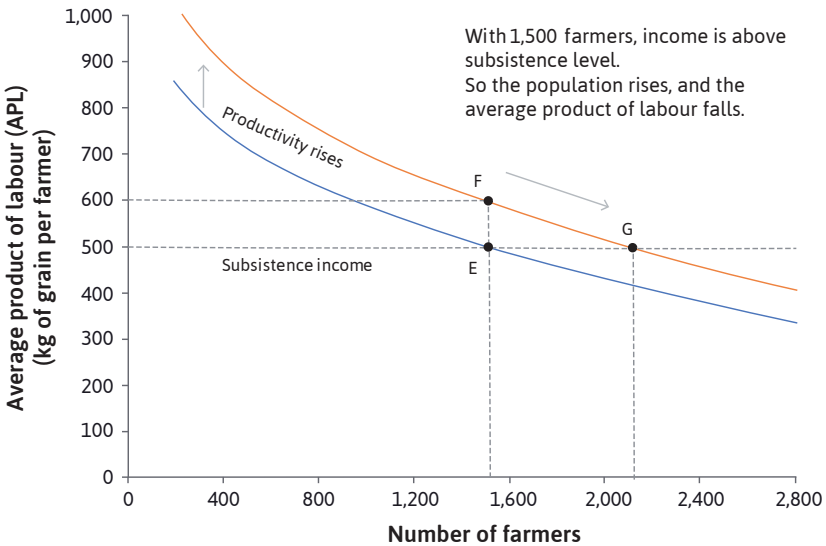


Figure 1.11 Malthus's model: the effect of an improvement in technology.

1. Initial equilibrium

The economy is initially in equilibrium at point E, with 1,500 farmers and subsistence income of 500 kg of grain per farmer.

2. An improvement in technology

When productivity rises, each of the 1,500 farmers produces more: income rises to 600 kg at point F.

3. A new equilibrium

Now that income is above subsistence level, the population begins to rise. And as the number of farmers increases, the average product of labour falls until a new equilibrium is reached at point G.

The Malthusian model predicts that improvements in technology will not raise living standards if:

- the average product of labour diminishes as more labour is applied to a fixed quantity of land
- population grows in response to an increase in living standards.

Then in the long run, an increase in productivity will result in a larger population but not higher living standards. This depressing conclusion was once regarded as so universal and inescapable that it was called Malthus's Law.

So we now have a possible explanation of the long, flat portion of the hockey stick. Human beings periodically invented better ways of making things, both in agriculture and in industry, and this periodically raised the incomes of farmers and employees above subsistence. The Malthusian interpretation was that higher incomes led couples to marry earlier and have more children, and they also led to lower death rates. Population growth eventually forced incomes back to subsistence levels, which might explain why China and India, with relatively sophisticated economies at the time, ended up with large populations but—until recently—very low incomes.

Looking for evidence

Can we find evidence to support the central prediction of the Malthusian model, that incomes will return to subsistence level?

Figure 1.12 is consistent with what Malthus predicted. Each point shows the population, together with an index of average wages—representing living standards—in England for a decade between 1280 and 1800. The index measures wages relative to the average wage in 1860.

In 1280 the population was just under five million, and the wage index stood at 61 (that is, 61% of the wage in 1860). Then in the fourteenth century, as a result of the outbreak of bubonic plague known as the Black Death, the population fell dramatically. The decline in population and labour supply had an economic benefit for the farmers and workers who survived: it meant that farmers had more and better land, and workers could demand higher wages. Incomes rose as the plague abated. Work through the figure to see what happened next.

The Black Death (https://tinyco.re/8267483) swept through Asia and Europe in the fourteenth century. It spread along trade routes, reaching Europe in the 1340s. An estimated 25 million people died: almost a third of the European population.

The seventeenth century was a period of technological progress in English agriculture. New crops (including potatoes), new techniques (such as the Dutch system of crop rotation), and new equipment (like the Chinese plough) raised productivity. Figure 1.12 shows the corresponding rise in incomes. But, just as the model in Figure 1.11 predicts, the improvement in living standards was only temporary. Population rose further, and wages began to fall. By the time Malthus was studying the problem in the 1790s, wages were below the average level for the previous five centuries.

EXERCISE 1.6 What would you add?

The Malthusian model makes use of many assumptions to analyse the effect of an improvement in technology.

1. How does this model simplify reality?
2. What has been left out?
3. Explain how including other factors that you think are important might change your analysis of how technological improvement affects the economy.

QUESTION 1.6 Choose the correct answer(s)

Figure 1.12 shows the wage index and population in England between 1280 and 1860.

- ☐ According to the Malthusian model, the fall in the population due to the bubonic plague would have led to an increase in the average productivity of workers, causing the observed rise in the wage index, post-plague.
- ☐ The doubling and halving of the wage index over 250 years from around 1350 is contrary to the Malthusian model.
- ☐ The data shows that technological improvements that occurred during the entire period shown had no effect on equilibrium.
- ☐ The data over this entire period is consistent with the conclusion of Malthus's model that a sustained increase in income per capita is not possible.

Robert C. Allen. 2001. 'The Great Divergence in European Wages and Prices from the Middle Ages to the First World War'. *Explorations in Economic History* 38 (4): pp. 411–447.

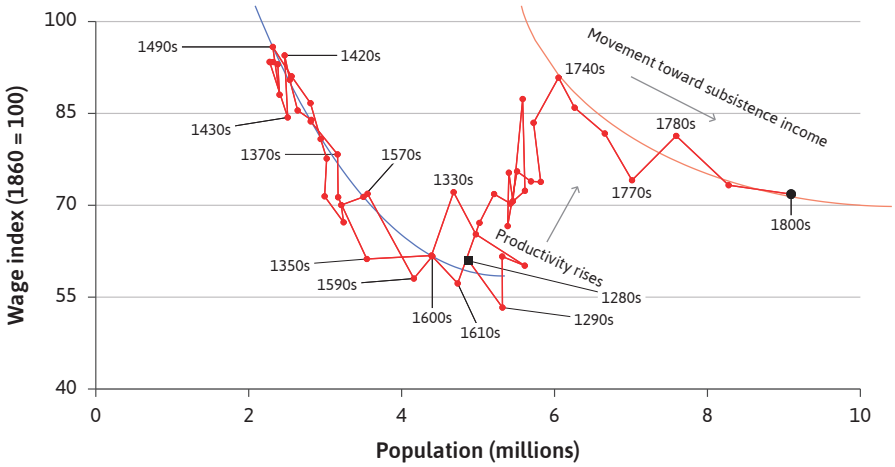


Figure 1.12 The Malthusian trap: wages and population (1280s–1800s)

1. Living standards and population over 500 years

Each point in the figure represents the wage index and population for a decade between the 1280s and the 1800s.

2. Between 1280 and 1490

Population fell in the fourteenth century, due to the plague, and wages rose. In the fifteenth century the population was only 2.3 million, and the wage index reached 96 in the 1490s.

3. Between 1500 and 1610

But Malthus's law asserted itself: higher wages helped the population to recover and, as it grew, wages fell. By 1610, population and living standards were close to the levels in 1280.

4. The effect of technological progress in the seventeenth century

Improving agricultural productivity during the seventeenth century resulted in higher wages from the 1620s onwards. The wage index reached 91 in the 1740s.

5. Another Malthusian trap

But again consistent with Malthus's model, higher incomes (indicated by the wage index) led to rising population, and as it rose, wages fell.

6. Malthus's Law holds

As predicted by Malthus's model (Figure 1.11), an increase in productivity resulted in a larger population in the 1800s, but not higher living standards.

1.8 Capitalist institutions

How can we explain the change from a world in which living conditions changed little unless there was an epidemic or a war, to the world of continuous technological revolution—in which each generation was noticeably, and predictably, better off than the previous one?

An important part of our answer is what we call the capitalist revolution: the emergence in the eighteenth century and eventual global spread of a way of organizing the economy that we now call capitalism. The term 'capitalism' was barely heard of a century ago, but Figure 1.13 illustrates how its use has skyrocketed since then. It shows the fraction of articles in the *New York Times* (excluding the sports section) that include the word 'capitalism'.

In everyday language, the word 'capitalism' is used in different ways, partly because people have strong feelings about it. In the language of economics, we use the term in a precise way: we define **capitalism** as an **economic system** characterized by a particular combination of **institutions**. An economic system is a way of organizing the production and distribution of goods and services in an entire economy. And by institutions, we mean the various sets of laws and social customs regulating production and distribution.

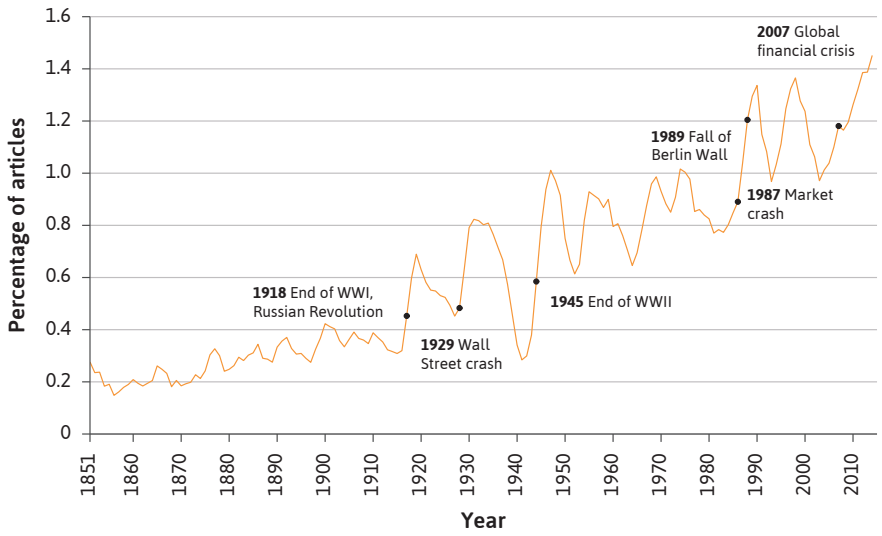
The institutions that characterize capitalism are private property, markets, and firms.

Since most of us live in capitalist economies, it is easy to overlook the importance of institutions that are fundamental for capitalism to work well. They are so familiar that we hardly notice them.

capitalism An economic system in which the main form of economic organization is the firm, where the private owners of capital goods hire labour to produce goods and services to be sold in markets with the intent of making a profit. The main economic institutions in a capitalist economic system are private property, markets, and firms.

economic system A way of organizing the economy that is distinctive in its basic institutions. Economic systems of the past and present include: central economic planning (e.g. the Soviet Union in the twentieth century), feudalism (e.g. much of Europe in the early Middle Ages), slave economy (e.g. the US South and the Caribbean plantation economies prior to the abolition of slavery in the nineteenth century), and capitalism (most of the world's economies today).

institution An institution is a set of laws and informal rules that regulate social interactions among people, and between people and the biosphere; sometimes also termed 'the rules of the game'.



Explore this data at https://tinyco.re/2871984

Calculations by Simon DeDeo, Santa Fe Institute, from *New York Times*. 2016. NYT article archive (https://tinyco.re/5093321).

Figure 1.13 Mention of the word 'capitalism' in *New York Times* articles (1851–2015).

Before seeing how these institutions combine in the capitalist economic system, we need to define them.

Over the course of human history, the extent of **private property** has varied. In some societies, such as that of our distant ancestors, the hunter-gatherers, almost nothing except personal ornaments and clothing was owned by individuals. In others, crops and animals were private property, but land was not. The right to use the land was granted to families by consensus among members of a group, or by a chief, without allowing the family to sell the plot.

In various economic systems, some human beings—as enslaved people—could be private property.

Private property may be owned by an individual, a family, a business, or some entity other than the government. Some things that we value are not private property: for example, the air we breathe and most of the knowledge we use cannot be owned or bought and sold.

PRIVATE PROPERTY
The right to private property means that you can:

- enjoy your possessions in a way that you choose
- exclude others from their use if you wish
- dispose of them by gift or sale to someone else (who then becomes their owner)

Markets are a means of transferring goods or services from one person to another. There are other ways, such as by theft, a gift, or a government order. **Markets** differ from these in three respects.

- Reciprocity. The transfer of a good on a market is reciprocated: unlike gifts and theft, one person's transfer of a good or service to another is directly matched by a transfer in the other direction (either of another good or service as in barter exchange, or money, or a promise of a later transfer when buying on credit).
- Participation is voluntary. All transfers are willingly undertaken, because the things being exchanged are private property and both buyer and seller expect to benefit: there are mutual **gains from exchange**.
- Competition. In most cases buyers and sellers are limited in the prices they can set. A seller charging a high price, for example, will find that buyers will purchase from other competing sellers offering lower prices.

MARKETS
Markets are a way of connecting people who may mutually benefit by exchanging goods and services through a process of buying and selling.

gains from trade, gains from exchange The benefits that each party gains from a transaction compared to how they would have fared without the transaction.

But private property and markets alone do not define capitalism. In many places they were important institutions long before capitalism. The most recent of the three components making up the capitalist economy is the **firm**.

The kinds of firms that make up a capitalist economy include restaurants, banks, large farms that pay others to work there, industrial establishments, supermarkets, and internet service providers. Other productive organizations that are not firms, and which play a lesser role in a capitalist economy, include family businesses, in which most or all of the people working are family members; non-profit organizations; employee-owned cooperatives; and government-owned entities (such as railways and power or water companies). These are not firms, either because they do not make a profit, or because the owners are not private individuals who own the assets of the firm and employ others to work there.

Figure 1.14 shows that the three parts of the definition of a capitalist economic system are nested concepts. The left-hand circle describes an economy of isolated families who own their tools and the goods they produce, but have little or no exchange with others.

Historically, economies like the left-hand circle have existed, but have been much less important than a system in which markets and private property are combined (the middle circle). Private property is an essential condition for the operation of markets: buyers will not want to pay for goods unless they can have the right to own them. In the middle circle, most production is done either by individuals (shoemakers or blacksmiths, for example) or in families (for example, on a farm) rather

than by firms with owners and employees. Prior to 1600, most of the economies of the world worked like this. Where firms, meaning employment for wages, existed, they played only a minor role.

In a capitalist system, which in Britain expanded during the seventeenth and later centuries, most production takes place in firms. Firms combine inputs—including equipment, raw materials, and labour—to produce outputs. Markets and private property are essential for firms to function for two reasons.

- *Outputs and capital goods are private property*: The firm's buildings, equipment, and other durable inputs, which are called **capital goods**, belong to the owners; so do the resulting outputs.
- *Firms use markets to buy inputs and sell outputs*: The owners' profits depend on markets in which customers may willingly purchase the products at a price that will more than cover production costs. And they buy their inputs in other markets, including the **labour market**, where they hire workers and pay them wages to operate their capital goods.

In other societies, private firms and markets have been relatively unimportant. The government has been the institution that owns capital goods, thereby controlling production, and deciding how goods should be distributed, and to whom. This is called a centrally planned economic system. It existed, for example, in the Soviet Union, East Germany, and many other eastern European countries prior to the end of Communist Party rule in the early 1990s.

Another contrast is with a slave economy: an economic system where most of the work is done by people who are not hired for wages but, instead, like the land on which they work, are the property of another person.

Though governments and families are essential parts of the workings of every economy, most economies today are capitalist according to our definition: they share the three characteristics of private property, markets, and firms. How these institutions of capitalism combine with each other and with families, governments, and other institutions differs greatly across countries—so it would be more precise to describe capitalism not as a specific economic system, but as a class of systems.

FIRMS
A firm is a way of organizing production with the following characteristics:

- One or more individuals own a set of capital goods that are used in production.
- They pay wages and salaries to employees.
- They direct the employees (through the managers they also employ) in the production of goods and services.
- The goods and services are the property of the owners.
- They sell the goods and services on markets with the intention of making a profit.

capital goods, capital Capital goods (sometimes shortened to 'capital') are the durable and costly non-labour inputs used in production (e.g. machinery, equipment, buildings). They do not include some essential inputs (e.g. air, water, knowledge) that are used in production at zero cost to the user.

labour market The market in which employers offer wages to individuals who may agree to work under their direction. Economists say that employers are on the demand side of this market, while employees are on the supply side.

Figure 1.14 Capitalism: private property, markets, and firms.

Capitalism, power, and competition

Capitalism is an economic system that combines centralization with decentralization. It concentrates power in the hands of owners and managers of firms who are then able to secure the cooperation of large numbers of employees in the production process. But it limits the powers of owners and of other individuals, because they face competition to buy and sell in markets.

So when the owner of a firm interacts with an employee, the owner is 'the boss'. But when the same owner interacts with a potential customer they are simply another person trying to make a sale, in competition with other firms. It is this combination of competition among firms, and concentration of power and cooperation within them, that accounts for capitalism's success as an economic system.

A striking characteristic of firms, distinguishing them from families and governments, is how quickly they can be born, expand, contract, and die. A successful firm can grow from just a few employees to a global company with hundreds of thousands of customers, that employs thousands of people, in a few years. Firms can do this because they are able to hire additional employees on the labour market, and attract funds to finance the purchase of the capital goods they need to expand production.

Firms can die in a few years too. A firm that does not make profits will not have enough money (and will not be able to borrow money) to continue employing and producing. The firm shrinks, and some of the employees lose their jobs.

Contrast this with a successful family farm. The family will be better off than its neighbours; but unless it turns the family farm into a firm, and employs other people to work on it, expansion will be limited. If, instead, the family is not very good at farming, then it will simply be less well off than its neighbours. The family head cannot dismiss the children as a firm might get rid of unproductive workers. As long as the family can feed itself, there is no equivalent mechanism to a firm's failure that will automatically put it out of business.

Government bodies also tend to be more limited in their capacity to expand if successful, and are usually protected from failure if they perform poorly.

How could capitalism lead to growth in living standards?

The emergence of capitalism brought two major changes, both of which enhanced the productivity of individual workers.

Technology

As we have seen, the continuous technological revolution coincided with the transition to firms as the predominant means of organizing production. This does not mean that firms necessarily caused technological change. But firms competing with each other in markets had strong incentives to adopt and develop new and more productive technologies, and to invest in capital goods that would have been beyond the reach of small-scale family enterprises.

Specialization

The growth of firms employing large numbers of workers—and the expansion of markets linking the entire world in a process of exchange—allowed historically unprecedented specialization in the tasks and products on which people worked. As Adam Smith explained, organizing production so that workers specialize in different tasks enables firms to increase output with the same number of employees.

Adam Smith also recognized some drawbacks of specialization: 'The man whose whole life is spent in performing a few simple operations, has no occasion to exert his understanding or to exercise his invention. He naturally loses, therefore, the habit of such exertion, and generally becomes as stupid and ignorant as it is possible for a human creature to become.'

You can read about his concerns in the article, 'Adam Smith's Two Views on the Division of Labour' (https://tinyco.re/6547660).

EXERCISE 1.7 Capitalism

Figure 1.13 (page 25) shows the proportion of *New York Times* articles that mention the word 'capitalism' (from 1851 to 2015).

1. Can you suggest an explanation for why the usage of the term capitalism spikes when it does?
2. Why do you think mentions of the word 'capitalism' have remained so high since the late 1980s?

EXERCISE 1.8 The poorest man's cottage

'The poorest man may in his cottage bid defiance to all the forces of the Crown. It may be frail, its roof may shake; the wind may blow through it; the storms may enter, the rain may enter—but the King of England cannot enter; all his forces dare not cross the threshold of the ruined tenement.' – William Pitt, First Earl of Chatham, speech in the British Parliament (1763)

1. What does this tell us about the meaning of private property?
2. To what extent does it apply to people's homes in your country?

EXERCISE 1.9 Markets and social networks

Think about a social networking site that you use, for example Facebook (https://tinyco.re/5641601). Now consider our definition of a market.

What are the similarities and differences between that social networking site and a market?

QUESTION 1.7 Choose the correct answer(s)

Which of the following are examples of markets?

- ☐ wartime food rationing
- ☐ auction websites such as eBay
- ☐ touts selling tickets outside concert halls
- ☐ sale of illegal arms

EXERCISE 1.10 Firm or not?

Using our definition, explain whether each of the following entities is a firm by investigating if it satisfies the characteristics that define a firm. Research the entity online if you are stuck.

1. John Lewis Partnership (UK)
2. a family farm in Vietnam
3. your current family doctor's office or practice
4. Walmart (US)
5. an eighteenth-century pirate ship (see our description of *The Royal Rover* in Unit 5 (page 0))
6. Google (US)
7. Manchester United plc (UK)
8. Wikipedia

1.9 Structural transformation: From farm to firm

With the growth of capitalist firms exploiting the opportunities of the technological revolution to raise productivity through innovation, the structure of the economy changed. Agriculture was no longer the primary source of income and output.

First, although agricultural output grew substantially, manufacturing output (and later, services) grew even faster and began to account for a much larger fraction of output. Second, partly as a result, the proportion (or share) of the population engaged in agricultural production fell while the manufacturing (and later, services) shares grew.

Figure 1.15 shows this structural transformation of the economy in several countries over 800 years. In all countries, especially in the last century and a half, the non-agricultural share of the workforce has risen. But the agricultural share in India and China remains high compared with the other economies. The economist, Arthur Lewis, developed a model of the economy that explains how this transformation takes place.

GREAT ECONOMISTS

Arthur Lewis

Arthur Lewis (1915–1991) won a Nobel Prize and received a knighthood from the British queen for his work in economics, but this was not his initial field of choice. 'I wanted to be an engineer,' he wrote, 'but this seemed pointless since neither the government nor the white firms would employ a black engineer.'

Born in the Caribbean island (and then British colony) of Saint Lucia, Lewis finished secondary school at age 14. As a young boy, required to remain at home for a couple of months due to illness, he learned so much studying by himself and being taught by his father that when he returned to school, he was transferred from the fourth to the sixth grade. At the London School of Economics, he was the student of another great economist, Friedrich Hayek (see Section 8.1 (page 0)).

What is now called the Lewis model represents an economy that has both a modern 'capitalist' sector producing industrial goods and a 'subsistence' sector of family farms producing just enough to sustain the family. The capitalist sector is composed of businesses, in which the owners profit from introducing new technologies to reduce the cost of the products that their employees produce. If the capitalist sector is profitable, the businesses expand, drawing in additional labour from the subsistence sector.

We use the Lewis model in the macroeconomics volume to explain why the hockey stick has not yet materialised in some countries.

Two aspects of the Lewis model explain the hockey stick. The first is the constant search for new technologies in the capitalist sector that allows more goods to be produced with an hour of labour, and thereby increase the business owners' profits. The second is that the capitalist sector is more productive than the subsistence sector (more output per hour) so moving workers out of the subsistence sector (illustrated by the 'non-agricultural share of the labour force' in Figure 1.15) increases the average productivity of the economy. And this raises income per capita as shown by the hockey stick in Figure 1.1 (page 4).

The essay that won him the Nobel Prize was titled 'Economic Development With Unlimited Supplies of Labour', referring to the fact that before the hockey stick tipped upwards, the subsistence sector covered most of the economy, so that the capitalist sector could expand by drawing in additional labour almost without limit. In that paper, Lewis drew attention to the fact that 'The classics, from [Adam] Smith to [Karl] Marx enquired how production grows through time [and] determined simultaneously income distribution and income growth …' He sought to return the attention of contemporary economists to these questions of growth and income distribution.

Arthur Lewis. 1954. 'Economic Development With Unlimited Supplies of Labour.' *Manchester School* 22: pp. 139–191.

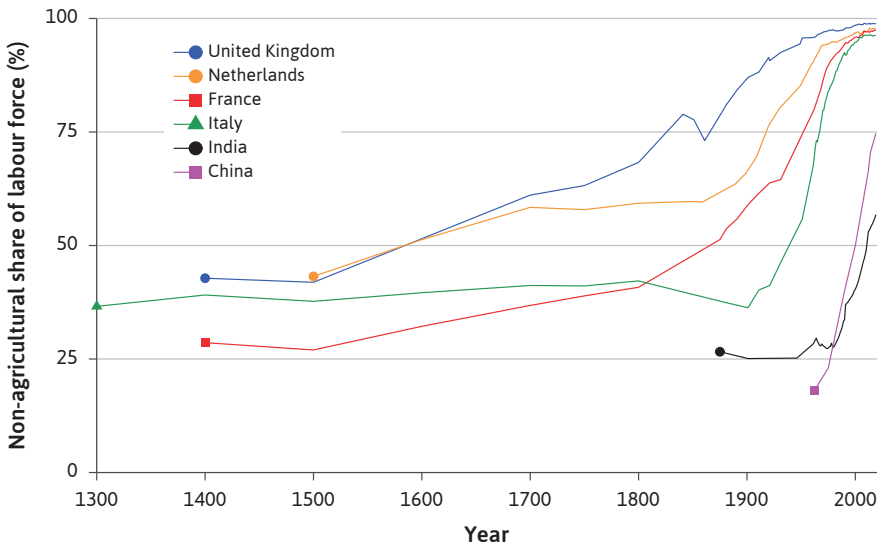
QUESTION 1.8 Choose the correct answer(s)

Read the following statements about Arthur Lewis and choose the correct option(s).

- ☐ Arthur Lewis investigated questions that were asked by the first economists.
- ☐ The Lewis model assumes that the supply of labour to the capitalist sector is unlimited.
- ☐ In the Lewis model, income per capita rises when workers move from the subsistence sector to the capitalist sector.
- ☐ One aspect of the Lewis model is the profit-seeking motives of the capitalist sector.

1.10 Capitalism, causation, and history's hockey stick

We have seen that the institutions associated with capitalism have the potential to make people better off, through opportunities for specialization and introduction of



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Tirthankar Roy. 2005. 'Agricultural Labor and Economic Transition in Colonial India: Lessons from Wage Data' (https://tinyco.re/3098756). Working Paper. 'World Development Indicators' (https://tinyco.re/2290878).; Pedro Lains and Vincente Pinella. 2009. 'Agriculture and Economic Development in Europe Since 1870' (https://tinyco.re/2075676). Routledge: Taylor and Francis India

Figure 1.15 Share of non-agricultural labour force in total labour force (1300–2019).

new technologies, and that capitalism emerged at the same time as, or just before, the continuous technological revolution. But can we conclude that capitalism caused the upward kink in the hockey stick? The emergence in the seventeenth century of a free-thinking cultural environment known as 'The Enlightenment' (https://tinyco.re/4014913) also predated or coincided with the upturn. So was it institutions, or culture, both, or some other set of causes? Economists and historians disagree, as we show in Unit 2 (page 0).

We want to make **causal** statements in economics—to understand why things happen, or to devise ways of changing something so that the economy works better. For example, an economist might claim: 'If the central bank lowers the interest rate, more people will buy homes and cars.'

But an economy is made up of the interactions of millions of people. We cannot measure and understand them all, and we should be sceptical when anyone claims that something complex (capitalism) 'causes' something else (increased living standards, technological improvement, or environmental challenges).

In science, we support the statement that X causes Y by performing experiments in which we change X and measure any associated change in Y. We design the experiments in such a way that we can be confident that the change in Y was caused by the change in X (rather than vice versa, or by some other variable). For most important questions in economics, it is not possible to conduct conventional experiments (although Unit 4 includes some examples where this is done). So how can economists study causation? The following example of a **natural experiment** shows how what we observe in the world can help us investigate causes and effects.

causal, causality, causation We can say that a relationship between two variables is causal if we can establish that a change in one variable produces a change in the other. While a correlation is simply an assessment that two things have moved together, causation implies a mechanism accounting for the association, and is therefore a more restrictive concept. *See also: natural experiment, correlation.*

natural experiment An empirical study that exploits a difference in the conditions affecting two populations (or two economies), that has occurred for external reasons: for example, differences in laws, policies, or weather. Comparing outcomes for the two populations gives us useful information about the effect of the conditions, provided that the difference in conditions was caused by a random event. But it would not help, for example, in the case of a difference in policy that occurred as a response to something else that might affect the outcome.

HOW ECONOMISTS LEARN FROM FACTS

Do institutions matter for growth in income?

In a conventional experiment, the experimenter changes the variable thought to be a cause (X), keeping all the other variables the same as before, and observes the change in Y that is associated with the change in X. In an agricultural experiment, X might be the dosage of fertiliser applied, the things kept the same (or 'held constant') might be the amounts of water and light, and Y would be the amount of the crop produced.

By contrast, in what is called a natural experiment, we rely on accidents of history, geography or other differences to 'perform the experiment' and to present us with different values of X that can then be associated with differing values of Y, when not much else that could cause changes in Y differs.

A natural experiment is a situation where we use an external event, such as a change in institutions or a natural disaster, to compare the outcomes for economic actors who were affected by this event and those who were not.

The division of Germany at the end of the Second World War into two separate economic systems—centrally planned in the east, capitalist in the west—provided a natural experiment. During this time a political 'Iron Curtain', as the British prime

More detail about Winston Churchill's 'Iron Curtain' speech (https://tinyco.re/6053919).

minister Winston Churchill described it, divided the country. It separated two populations that until then had shared the same language, culture, and capitalist economy. But for half a century they differed in the institutions governing the economy. This allows us to study the effect of different institutions on the economy.

Using the language of experiments, central planning under Communist Party rule in East Germany is the 'treatment' and capitalism in West Germany is the 'control' that provides an indication of how the East would have developed had Germany not been divided. The control is also termed the 'counterfactual' event: what would have happened if, contrary to the facts of history, East Germany had not been organized as a planned economy between 1948 and 1991.

The validity of this as a natural experiment relies on the comparability of the two parts of Germany before the introduction of different economic systems. In 1936, before the Second World War, living standards in both parts of Germany were the same. Firms in Saxony and Thuringia were world leaders in automobile and aircraft production, chemicals, optical equipment, and precision engineering. During the war, the entire German economy was organized as a war economy: for example, wages and prices and the allocation of labour and resources were under central control.

After the war, Communist central planning replaced the war economy in East Germany, where the Soviet Union was the occupying power. Decisions about how and what to produce did not revert to private individuals, but were taken by government officials. The officials managing plants, offices, mines, and farms did not need to follow the principle of capitalism and produce goods and services that customers would buy at a price above their cost of manufacture. Pre-war private property, markets, and firms virtually disappeared.

But a capitalist economy re-emerged in West Germany as a result of decisions by the occupying powers (the UK, US, and France). Following currency reform in 1948, which abolished wartime price controls and introduced a new currency, rapid growth occurred.

The East German Communist Party forecast in 1958 that material wellbeing there would exceed that of West Germany by 1961. The failure of this prediction was one of the reasons the Berlin Wall separating East from West Germany was built in 1961. By the time the Wall fell in 1989, and East Germany abandoned central planning, its GDP per capita was less than half of that of capitalist West Germany. Figure 1.16 shows the different paths taken by these and two other economies from 1950.

Figure 1.16 shows that West Germany started from a more favourable position in 1950 than East Germany. In 1936, before the war began, the two parts of Germany had virtually identical living standards; both had achieved successful industrialization. East Germany's relative weakness in 1950 was not mainly because of differences in capital equipment or skills, but because its industrial structure was more disrupted by splitting the country than was the case in West Germany.

You can read more examples of natural experiments in the book *Natural Experiments of History* (https://tinyco.re/8903951), written by Jared Diamond, a biologist, and James Robinson, a professor of government.

Explore this data at https://tinyco.re/6997062

The Conference Board. 2015. *Total Economy Database* (https://tinyco.re/1587851). Angus Maddison. 2001. 'The World Economy: A Millennial Perspective'. Development Centre Studies. Paris: OECD.

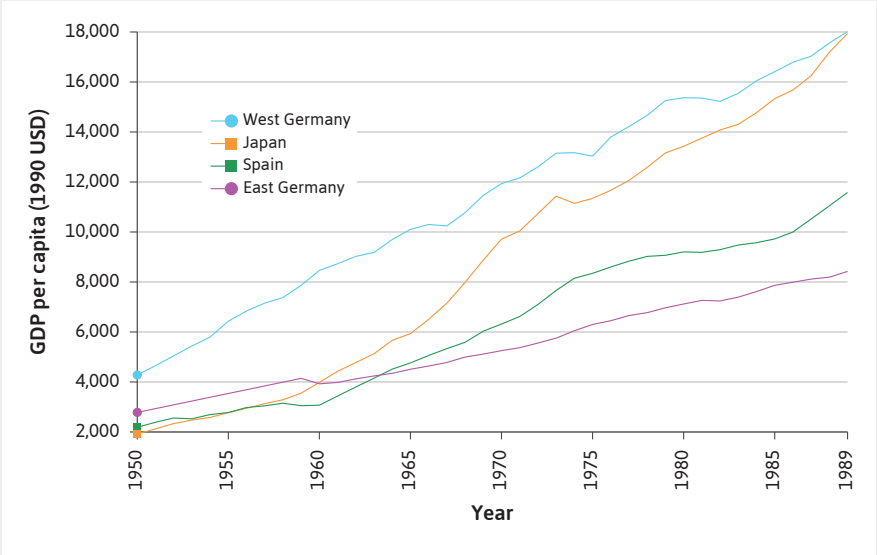

Figure 1.16 The two Germanies: planning and capitalism (1950–89).

But other capitalist economies that had even lower per capita incomes in 1950, grew much faster. By 1989, the Japanese economy, which had also suffered war damage, had—with its own combination of private property, markets, and firms, along with strong government coordination—caught up to West Germany. By 1989, Spain, which until 1975 like East Germany was ruled by a dictatorship, had closed part of the gap. Growth accelerated after the market reforms introduced in 1959.

We cannot conclude from the German natural experiment that capitalism always promotes rapid economic growth, while central planning is a recipe for relative stagnation. What we can infer is more limited: during the second half of the twentieth century, the divergence of economic institutions mattered for the material living standards of the German people. For a longer run view of comparative performance under central planning (the Soviet Union and Latin American countries), see Figure 1.19 (page 40) in Section 1.12.

QUESTION 1.9 Choose the correct answer(s)

Figure 1.16 shows a graph of GDP per capita for West and East Germany, Japan, and Spain between 1950 and 1990. Read the following statements and choose the correct option(s).

- ☐ Having a much lower starting point in 1950 was the main reason for East Germany's poor performance compared to West Germany.
- ☐ The fact that Japan and West Germany have the highest GDP per capita in 1990 implies that they found the optimal economic system.
- ☐ Spain was able to grow at a higher growth rate than Germany between 1950 and 1990.
- ☐ The difference in East and West Germany's performance proves that capitalism always promotes rapid economic growth while central planning is a recipe for stagnation.

1.11 Application: Did the British colonization of India reduce Indian living standards?

The technological revolution and trade reversed the fortunes of the British and Indian textile industries.

British colonization of India began in 1600, and over the next 200 years Britain became the dominant European colonial power there, taking full control by the mid-nineteenth century.

A prelude to the Industrial Revolution in Britain was a period of booming sales of novel types of clothing made in India using traditional technology. 'All of a sudden, we saw all our women, rich and poor, clothed in [India-made] calico,' wrote Daniel Defoe, author of Robinson Crusoe, in 1727. He went on: 'China, India, and other Eastern Countries ... have, it is true, the most extended manufacture, and the greatest variety in the world, and their manufactures push themselves upon the world, by the mere stress of their cheapness.'

A century and a half later, the flow of trade had been reversed. Over half of the textiles purchased in India were made in Britain's steam-powered mills and the sub-continent's once world-dominant textile industry was in permanent decline. India had been one of the largest producers of manufactured goods in the world, but under British colonial rule would remain a farm-based economy. Following the decline of its textile industry, its main exports were agricultural products, until late in the twentieth century, when India became a major exporter of software and computer services.

Angus Deaton, a Nobel Laureate who specializes in the analysis of poverty, observed that, when 300 years of British rule of India ended in 1947, 'It is possible that the deprivation in childhood of Indians ... was as severe as that of any large group in history'. In the closing years of British rule, a child born in India could expect to live for 27 years. After fifty years of independence, life expectancy at birth had risen to 63 years.

Figure 1.17 (an enlargement of part of Figure 1.1 (page 4)) shows that per capita output declined in both India and China in the 250 years before European colonial powers were heavily involved in these countries. The take-off of the Indian economy began soon after independence from colonial rule in 1946. In China, civil conflict during the Ming-Qing transition during the 17th century wiped out about a quarter of the population, raising per capita income, while the subsequent recovery and acceleration of population growth reversed thus, reducing agricultural productivity

and per capita GDP during the 18th century. This Malthusian trap process is explained in Section 1.7. GDP per capita then fluctuated without trend during the 19th century.

The Indian data raises the question: 'Did the British colonization of India reduce Indian living standards?'

To answer this question fully, we would need to know what would have happened if India had remained independent rather than becoming a British colony. In the previous section, we addressed a similar type of question by using West Germany as a 'control' or 'counterfactual', to indicate what would have happened in East Germany had it not been a planned economy. Can we find an appropriate counterfactual for Indian history without British rule?

Japan might be a suitable comparator: it had been as poor as India in 1600 (Figure 1.1 (page 4)) but successfully resisted domination by any foreign power. From the late nineteenth century onwards, Japan became a successful capitalist economy that was adept at borrowing the new technologies developed in Britain and other economies that were already part of the continuous technological revolution. We cannot tell whether something similar would have happened in India had it remained independent.

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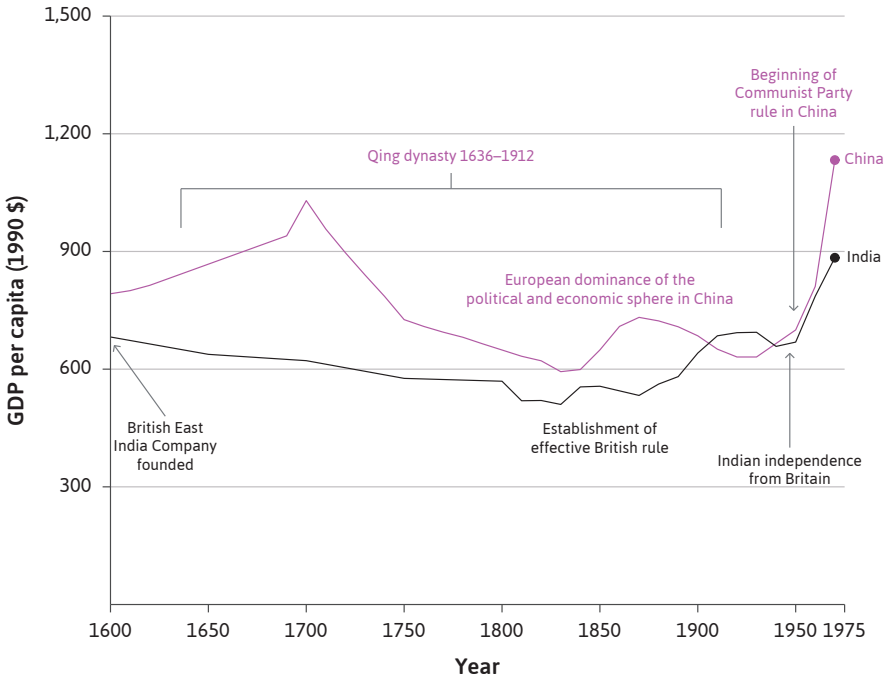


Figure 1.17 Periods of foreign intervention in India and China: Trends in average income. Note: Average incomes declined in India from 1600 and China from 1700. After foreign influence became dominant in the 19th century, living standards barely increased until the middle of the 20th century. By contrast, in Britain average income in 1975 was 11 times what it had been in 1600. Average income in Japan, a country that resisted foreign interventions, increased 17 times over the period shown in Figure 1.17. The growth of both the Indian and Chinese economies accelerated after 1975.

Recent research documents two of the actions taken by the British colonial rulers that appear to have reduced Indian living standards.

Railways for military use, not economic development: In India the railways were designed primarily for military purposes, particularly after the revolt of Indian soldiers in the British colonial army in 1857. By contrast, the growing prosperity of farming regions in the US, Canada, and other economies was supported by the development of public infrastructure, particularly railways and ports, to expand the markets for farm products and the development of industries processing them.

The transport infrastructure that served Indian agriculture was used to export agricultural products, not transport them to local manufacturers for processing. For example, hides and skins were exported rather than being locally manufactured into shoes. The result appears to have been a lost opportunity for the growth of the farm-based economy and industries using its output.

Entrenching local elites: In many parts of the subcontinent, British colonial rulers relied on local elites—particularly large landowners—to collect taxes for them (taking a cut for themselves, of course). The result was the entrenchment of the power and property rights of powerful landlords who were more interested in collecting taxes than in modernizing the economy. Their influence over those renting land from them or working for them for wages was already substantial before British rule, but it was greatly strengthened by the Permanent Settlement of 1793. In this agreement, the East India Company (the de facto colonial rulers) conferred governmental powers on the landlords by giving them the right to collect taxes and making them de facto state officials.

The fact that British taxation and land tenure policy varied across the colony provides a natural experiment to test the importance of these British institutions for subsequent economic growth. Economists Abhijit Banerjee and Lakshmi Iyer compared the post-independence economic performance and social indicators of districts of modern-day India in which landlords had been empowered by the colonial land tenure and taxation systems with other districts where the landlords had been bypassed in favour of the village community or direct taxation of the individual farmer. They found that the landlord-controlled districts had significantly lower rates of agricultural productivity growth, reflecting lower rates of investment and less use of fertilisers and other modern inputs. The landlord-controlled districts also lagged significantly in educational and health improvements.

Colonial rule of India coincided with the divergence of fortunes of Britain and India. The eclipse of India's early lead in textile manufacturing and its failure to transition away from a stagnating agricultural economy dominated by rural elites is part of the legacy of colonial rule there.

Economist and historian Bishnupriya Gupta explores the stagnation and decline of Indian living standards prior to Independence in 1947.

Professor Bishnupriya Gupta explains the long period of economic stagnation, and even decline, during British rule in India and considers why economic prosperity accelerated following Indian independence in 1947.

QUESTION 1.10 Choose the correct answer(s)
In the period between 1600 and 1750 (before the Industrial Revolution took off in Britain), India's GDP per capita was:

- ☐ declining, because agricultural productivity was declining, and most people were working in agriculture
- ☐ declining, because the formerly booming textile industry was being out-competed by British textiles
- ☐ rising, because the domestic textile industry was booming, and more people worked in industry than in agriculture
- ☐ closer to the level of British GDP per capita than it was two hundred years later between 1800 and 1950.

Though China was never a colony of a European nation, from the early nineteenth century Britain, Portugal, and other nations controlled a number of key ports, such as Hong Kong and Macau, and prevented the Chinese government from acting independently in important economic matters. These foreign interventions ended only in the mid-twentieth century, when a successful revolution ended foreign interventions and placed China under Communist Party rule. During this period of foreign intervention, per capita income in China declined.

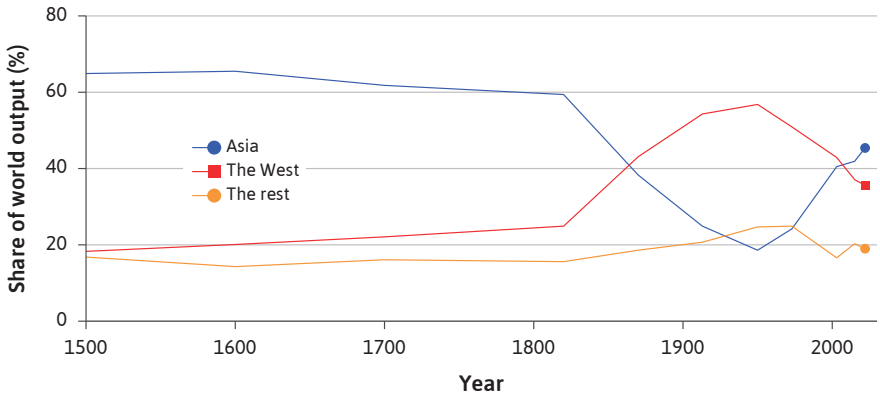
The rise of Britain and then the rest of Europe—as both economic and military powers—fundamentally altered the geography of worldwide production, that is, where most global goods and services (output) were being produced. If we measure the size of an economy by the total amount of goods and services (output) it produces, we see dramatic shifts since Tavernier visited India in the seventeenth century:

- In 1600, India, China, and the rest of Asia produced three-fifths of the world's output, three times as much as western Europe.
- In 1750, China produced a third of the world's manufacturing output and India's textile and other manufacturing output was greater than in the whole of Europe.
- And only in 1850 did London displace Beijing as the world's most populous city.

Figure 1.18 summarizes the changing geography of world output, showing the percentage of global output produced in three different regions.

For a visual reminder of these long-run changes, note that the time paths of the shares of world output of the West and Asia resemble a pair of pliers.

- Before 1800, Asia's large share of world output reflects the fact that most people in the world lived in Asia (the amount produced per person did not differ much by region).
- Beginning around 1800, there was a sharp rise in the contribution of the West to world output at the expense of Asia's contribution because the European population grew rapidly, as did output per person.
- In the last few decades, with the rapid growth of East and South Asia (especially India and China, but also other countries including Vietnam and South Korea), Asia is once again contributing a larger share of world output than the West.



Angus Maddison. 2007. Contours of the world economy 1–2030 AD: Essays in macro-economic history. OUP Oxford.; IMF World Economic Outlook (https://tinyco.re/5598070). 2022.

Figure 1.18 Shares of world output produced in Asia, the West, and the Rest. The West refers to western Europe, the US, Canada, Australia, and New Zealand. The Rest includes Latin America, Japan, Africa, and eastern Europe (the latter including Russia and other countries ruled by the Communist Party in the second half of the twentieth century).

QUESTION 1.11 Choose the correct answer(s)

Read the following statements and choose the correct option(s).

- ☐ In 1727, Daniel Defoe, the author of Robinson Crusoe, described how China, India, and other Eastern countries had very limited manufacturing activity.
- ☐ Research by economists Abhijit Banerjee and Lakshmi Iyer found that colonial era tax systems which empowered large landlords are associated with lower current-day levels of agricultural productivity, education, and health.
- ☐ In India, public infrastructure such as railways and ports was designed to expand the markets for farm products and the development of industries processing them.
- ☐ More than 60% of global GDP has been produced in the West (Europe, the US, Canada, and Australia) consistently from as early as 1500 until today.

Some researchers question the validity (https://tinyco.re/5201040) of historical GDP estimates like these outside of Europe, because the economies of these countries were so different in structure.

1.12 Varieties of capitalism: Institutions, government, and politics

Not every capitalist country is the kind of economic success story exemplified in Figure 1.1 (page 4) by Britain, later Japan, and the other countries that caught up. Figure 1.19 tracks the fortunes of a selection of other countries during the twentieth century. In Africa, the success of Botswana in achieving sustained growth contrasts sharply with Nigeria's relative failure to do so. Both are rich in natural resources (diamonds in Botswana, oil in Nigeria), but differences in the quality of their institutions—the extent of corruption and misdirection of government funds, for example—may help explain their contrasting trajectories.

The star performer in Figure 1.19 is South Korea. In 1950 its GDP per capita was the same as Nigeria's, but by 2013 South Korea was six times richer than Nigeria by this measure.

developmental state A government that takes a leading role in promoting the process of economic development through its public investments, subsidies of particular industries, education, and other public policies.

South Korea's take-off occurred under institutions and policies sharply different from those prevailing during the take-off in Britain. The most important difference is that the South Korean government (along with a few large corporations) played a leading role in directing the process of development, explicitly promoting some industries, requiring firms to compete in foreign markets and providing high-quality education for its workforce. The term **developmental state** refers to cases like South Korea, Japan, and China, where the government plays a leading role in economic take-off.

In 1928, GDP per capita in the Soviet Union was a third of that in Argentina, and higher than in Brazil and South Korea. Communist central planning produced steady growth for nearly 50 years, outstripping Brazil by a wide margin and even overtaking Argentina before Communist Party rule ended in 1990.

One reason why central planning was abandoned as an economic system was its failure—as illustrated by the case of East Germany—to deliver the improvements in living standards achieved by some capitalist economies. Yet the transition to a new economic system in the former Soviet Union following the collapse of communism proved very costly. Dismantling the planned economy and creating functioning market economy institutions, including the rule of law, in the successor economies was accompanied by a deeper and more prolonged decline in per capita output than observed in the Second World War, the global financial crisis, or the COVID-19 pandemic.

When is capitalism dynamic?

The lagging performances of some economies in Figure 1.19 demonstrate that the existence of capitalist institutions is not enough, in itself, to create a dynamic economy —that is, an economy that brings sustained growth in living standards. Two sets of

Explore this data at https://tinyco.re/2023925

Jutta Bolt and Jan Luiten van Zanden. 2020. Maddison Project Database, version 2020. 'Maddison style estimates of the evolution of the world economy. A new 2020 update' (https://tinyco.re/7763210).

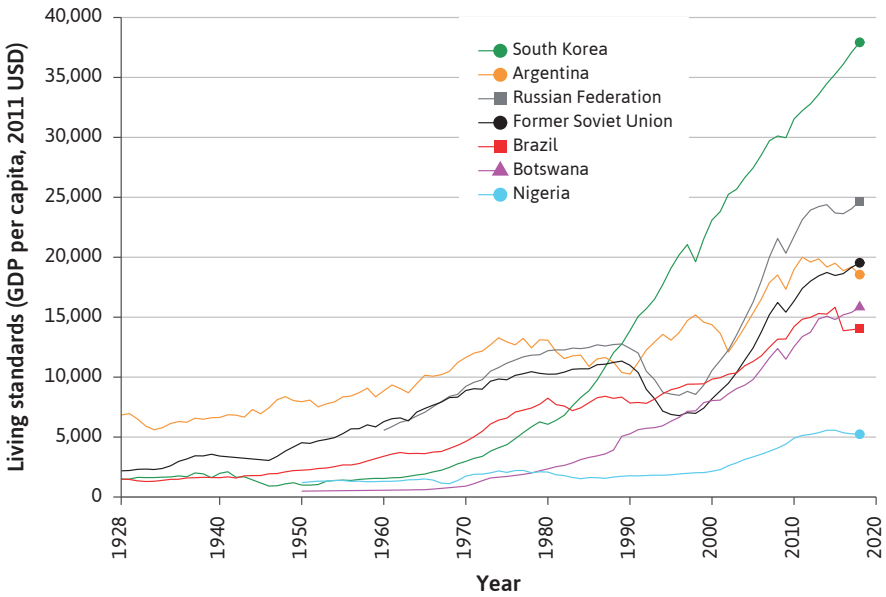


Figure 1.19 Divergence of GDP per capita among latecomers to the capitalist revolution (1928–2018).
Note: Former Soviet Union series excludes Russian Federation post-1992.

conditions affect the dynamism of the capitalist economic system. One set is economic; the other is political, and concerns the government and the way it functions.

Economic conditions

The potential advantages of capitalism will not be realized if:

- *Private property is not secure*: There is weak enforcement of the rule of law and of contracts, or expropriation either by criminal elements or by government bodies.
- *Markets for firms' output are not competitive*: Firms lack incentives to produce efficiently, expand, and innovate.
- *Firms are controlled by people who owe their positions to connections to government or privileged birth*: They did not become owners or managers because they were good at delivering high-quality goods and services at a competitive price. The other two failures would make this more likely to occur.

Failures of the three institutions of capitalism can mean that individuals and groups gain more from lobbying and criminal activity to shift the distribution of income in their favour, than from the direct creation of economic value.

Capitalism is the first economic system in history in which membership of the elite often depends on economic performance. A feudal lord who managed his estate poorly was just a shabby lord. But firm owners who fail go bankrupt—and a bankrupt owner is an ex-owner. The discipline imposed by market competition—produce and sell good products profitably, or fail—is automatic. Competition provides a mechanism for weeding out firms and individuals who underperform.

Of course, if the owners and managers of capitalist firms are initially wealthy or well connected politically, they may survive despite their failures. And, as Adam Smith pointed out, capitalism brings with it powerful incentives for firms to reduce the competition they face. But there are no guarantees: staying ahead of the competition means constantly innovating.

The Uber Files (https://tinyco.re/2355645): A leak of confidential documents from the ride-hailing company, Uber, shows how its expansion into cities across the world was fuelled not only by rapid technical and operational innovation, but also by aggressive (and sometimes illegal) competition, and constant political lobbying.

Political conditions

Government is also important. Irrespective of the extent of their direct involvement in the economy, governments establish, enforce, and change the laws and regulations that influence how the economy works. Markets, private property, and firms are all regulated by laws and policies.

A well-functioning legal system is essential for markets to work well. Governments adjudicate disputes over ownership and enforce property rights, and they establish the regulations that prevent firms from undermining market competition by colluding with each other or engaging in unfair competitive practices.

In addition to supporting the institutions of capitalism, the government provides essential goods and services such as physical infrastructure, education, and national defence. In this book, we investigate why government policies in such areas as sustaining competition, taxing, and subsidizing to protect the environment, thereby influencing the distribution of income, the creation of wealth, and the level of employment and inflation may also make good economic sense.

political system A political system determines how governments will be selected, and how those governments will make and implement decisions that affect all or most members of a population.
democracy A political system that ideally gives equal political power to all citizens, and which is defined by individual rights such as freedom of speech, assembly, and the press; and fair elections in which virtually all adults are eligible to vote, and the government leaves office if it loses.

Political systems

One of the reasons why capitalism comes in various forms is that it coexists with different political systems. A **political system**, such as **democracy** or dictatorship,

determines how governments are selected, and how those governments make and implement decisions that affect the population.

Capitalism emerged in Britain, the Netherlands, and most of today's high-income countries long before democracy. In no country were most adults eligible to vote before the end of the nineteenth century (New Zealand was the first). Even in the recent past, capitalism has coexisted with undemocratic rule, as in Chile from 1973 to 1990, Brazil from 1964 to 1985, and Japan until 1945. Contemporary China has a variant of capitalism with a high level of state intervention, but its system of government is not a democracy by our definition. In many countries today, however, capitalism and democracy coexist, each system influencing how the other works.

Like capitalism, democracy comes in many forms. The head of state may be elected directly by the voters, or by an elected body, such as a parliament. Some democracies strictly limit the ways in which individuals can influence elections or public policy through financial contributions; in others, private money has great influence through contributions to electoral campaigns, lobbying, and even bribery.

These differences even among democracies are part of the explanation for governments' differing roles in the capitalist economy. The Japanese and South Korean governments play a central role in setting the direction of the economy. But the amount of tax collected (both locally and nationally) is low compared with some rich countries in northern Europe, where it is almost half of GDP. In Sweden and Denmark, the tax system is used to reduce income inequality to a far greater extent than in Japan and South Korea.

QUESTION 1.12 Choose the correct answer(s)

Based on the information in Figure 1.19, read the following statements and choose the correct option(s).

- ☐ The Communist Party rule in the former Soviet Union before 1990 was a complete failure.
- ☐ The contrasting performances of Botswana and Nigeria illustrate that rich natural resources alone do not guarantee higher economic growth, but that higher-quality institutions (government, markets, and firms) may also be necessary.
- ☐ The impressive performance of South Korea's economy implies that other countries should copy their economic system.
- ☐ The evidence from the Russian Federation and the former Soviet Union after 1990 shows that the replacement of central planning by capitalism led to immediate economic growth.

1.13 Economics, the economy, and the biosphere

economics Economics is the study of how people interact with each other and with their natural environment in producing and acquiring their livelihoods, and how this changes over time and differs across societies.

We define **economics** as the study of how people interact with each other and with their natural environment in producing and acquiring their livelihoods, and how this changes over time and differs across societies.

Humans have always relied on their environment for the resources they need to live and produce their livelihoods: the physical environment and the biosphere (the collection of all forms of life on earth) provide essentials for life such as air, water, and food. The environment also provides the raw materials that we use in the production of other goods—such as wood, metals, and oil.

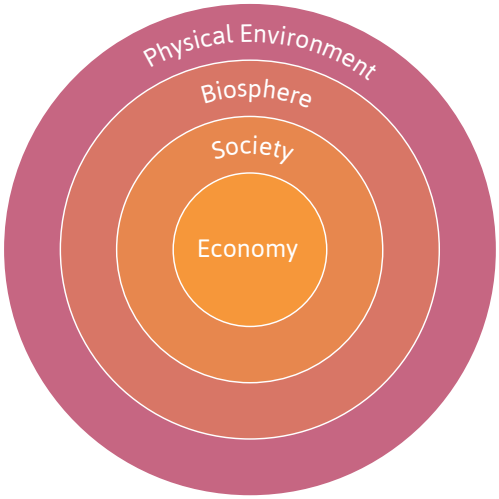


Figure 1.20 The economy is part of society, which is part of the biosphere.

Figure 1.20 shows one way to think about the economy: it is part of a larger social system, which is itself part of the biosphere. People interact with each other, and also with nature, in producing their livelihood.

Figure 1.21 shows the flows of resources between firms and households within the economy, and between the economy and the environment. Firms combine labour with structures and equipment, and produce goods and services that are used by households and other firms.

Production of goods and services also takes place within households although, unlike firms, households may not sell their outputs in the market. In addition to goods and services, households are also producing people—the next generation of the labour force. The labour of parents, caregivers, and others is combined with structures (for example, your home) and equipment (for example, the washing machine in the home) to reproduce and raise the future labour force working in firms, and the people who will work and reproduce in the households of the future.

Figure 1.22 shows that environmental resources are essential for sustaining both human life and the economic activity that supports and enhances it. The environment provides resources that we consume directly, and the primary inputs to the production of goods and services within households and firms.

According to the *Living Planet Report 2020* by the World Wildlife Fund, there was an average 68% decrease in population sizes of mammals, birds, amphibians, reptiles, and fish between 1970 and 2016. And since 1970, humanity's *ecological footprint* has exceeded the earth's rate of regeneration. The report estimates that we are overusing the earth's biocapacity by at least 56%. Figure 1.22 illustrates projections for future biodiversity depending on whether we can make human production and consumption more sustainable.

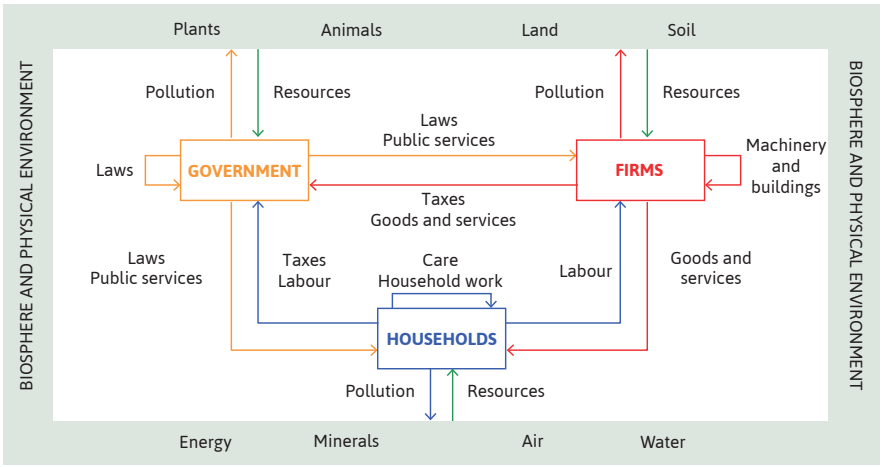


Figure 1.21 A model of the economy: flows of resources.

We have seen that capitalist institutions—private property, markets, and firms—have facilitated the continuous technological revolution and rapidly rising living standards by encouraging innovation and successful adoption of new technologies. But many of the most important technological developments of the twentieth century (https://tinyco.re/2009071), from cars and aeroplanes to refrigeration and personal computers, have relied on carbon-based energy for their use as well as manufacture. Others, like plastics and chemical fertilisers, lead directly to ecosystem damage.

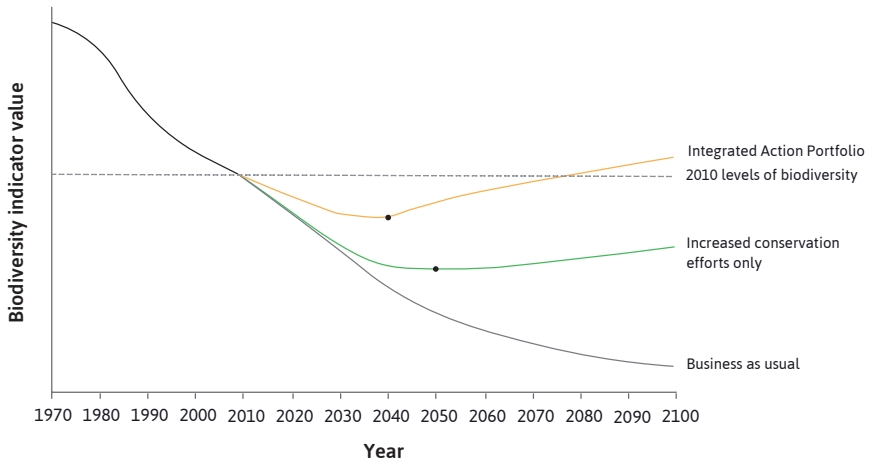


Figure 1.22 Global biodiversity loss under three different scenarios. (Integrated Action Portfolio refers to increased conservation efforts, increased agricultural land productivity, reductions in food waste, and changes in food consumption behaviour.) Labelled points indicate the year in which biodiversity is expected to start increasing.

The evidence on biodiversity and climate change shows how dramatically the way we produce our livelihoods is now degrading the environment, and depleting the stock of natural resources—including clean air and a liveable climate.

An example is the Grand Banks cod fishery off the east coast of Newfoundland, which sustained the livelihoods of the US and Canadian fishing communities for 300 years. Cod stocks collapsed in 1992, after several decades of large-scale commercial fishing, and the fishery closed. We still do not know if the cod will come back in their previous numbers.

One reason for the environmentally destructive pattern of technological change and resource use is that goods or technologies that deplete or destroy natural resources are artificially cheap, so we overuse them. They are artificially cheap because the prices that users of natural resources pay do not include the depletion of the natural environment. For example, what consumers pay when they buy cod goes into the wages of the crew and others employed in the supply chain, and into the profits of the firms involved. But no fishing firm owner has an incentive to maintain the stock of cod on the Grand Banks. The cod stocks could be freely exploited while they lasted; if the owner of a trawler tried to conserve them by catching fewer fish per week, their own profits would be reduced and competing trawlers would catch the fish anyway.

The depleted cod stocks parallels the case of the depletion of the Indonesian forests in Extension 1.2 (page 5), which is a cost of forestry there that is not deducted from GDP. In this case, the value of the fish sold is counted in GDP (of Canada, say), but the depletion of the stock of fish is not subtracted.

This contrasts with a firm owner whose profits depend on a stock of privately owned resources and who has an incentive to maintain it in good condition: for example, a tour operator will maintain their fleet of buses because they are needed tomorrow as well as today, and are costly to replace.

Some environmental problems can be addressed by governments directly regulating the amount of emissions or other environmental damages. Examples are banning lead in petrol (gasoline) or issuing a limited number of permits to emit CO_2 and allowing firms to buy and sell these permits. Without regulation, electricity generators do not pay for using up the absorptive capacity of the biosphere. By limiting the total number of permits issued, this policy limits the total amount of emissions and puts a price on the use of CO_2 because firms emitting it have to buy permits. This also provides a profit motive for owners of firms to reduce carbon emissions that is absent when they are not regulated. These policies protect the environment by making goods produced in environmentally harmful ways either illegal or costly, so they will be used less.

One way that technological progress can contribute to mitigating climate change and biodiversity loss is by reducing the cost of goods and services that are compatible with environmental sustainability. Recent advances in technology have vastly reduced the cost of wind, solar, and other renewable sources of energy. We discuss some examples in the next unit.

The evidence on biodiversity and climate change shows how dramatically the way we produce our livelihoods is now changing the environment, and depleting the stock of natural resources. By treating them as free, and using them up, we affect future (human) living standards and wellbeing. From global climate change to local resource exhaustion, these effects are results of both the expansion of the economy (illustrated by the growth in total output) and the way the economy is organized (what kinds of things are valued and conserved, for example).

Addressing environmental problems to enable those in low-income countries to transition out of poverty and to sustain living standards in rich ones requires governmental and other collective solutions, whether local, national, or international. Local communities can organize recycling schemes, or agree on

regulations for the use of a lake. Governments can limit or prevent the sale of damaging products, as some have done in the case of incandescent light bulbs or petrol- or diesel-powered cars; or subsidise beneficial investment, for example in public transport infrastructure, solar and wind power, or home insulation. Climate change, and the conservation of oceans and some rivers, requires not only individual government action, but international agreement.

Although governments are far more able than individuals and firms acting singly to take action to protect the environment, they often fail to do so. The centrally planned economies of Eastern Europe, where governments controlled production, had a particularly poor record on pollution control. Per capita mortality from air pollution in Eastern Europe (outside the EU) and China remains high relative to the EU and North America. But although democratic governments have been more active in reducing pollution that negatively affects the lives and health of their citizens, they have been reluctant to adopt environmental policies that restrict individual choice—for example, to tax or limit the use of private cars—or that would reduce profits of companies providing carbon-based energy.

EXERCISE 1.11 Earth Overshoot Day

Earth Overshoot Day, an initiative by the Global Footprint Network, marks the date when global demand for environmental and natural resources in a given year exceeds what the earth can regenerate in that year. In 2022, that day was July 28. However, each country uses the earth's resources at different rates.

1. Check the Earth Overshoot Day website's diagram (https://tinyco.re/9767876) of country-specific overshoot days to find out the date for your country (or the country you live in) in the latest year available.
2. Take this Ecological Footprint Calculator survey (https://tinyco.re/3239841) to find out your personal overshoot day (or equivalently, how many earths we would need if everyone lived a similar lifestyle to you).

1.14 Summary

- For most of human history, living standards remained at what we would today consider to be a low level with temporary ups and downs associated with variations in weather, disease, and social conflicts.
- Beginning in the 1700s in Great Britain, rising average material living standards have become a continuous feature of economic life in many countries; but one in ten people in the world still live in extreme poverty.
- The rising level of global output has contributed to climate change, biodiversity loss, and other threats to the environment, especially since the beginning of the twentieth century.
- Contributing to both rising affluence and environmental threats was the introduction of new methods of production, increasing the amount a person can produce in an hour by harnessing energy based on burning carbon.
- The technological revolution and resulting continuous increases in output were facilitated by a new economic system called capitalism, which combines three institutions: private property, markets, and firms hiring others to produce goods in order to make a profit.

- Differing forms of capitalism and differing political systems have led to differences in economic performance between countries.
- Many countries that were colonized, like India, did not experience substantially rising living standards until well after independence. For other countries, many in Latin America, for example, independence from colonial rule (in the early nineteenth century) did not bring about a change in economic fortunes.
- Economics is the study of how people interact with each other and with the natural environment in producing their livelihoods.
- Climate change and biodiversity loss mean that we need new non-carbon technologies and new policies and institutions to sustain our planet, and at the same time eliminate global poverty.
- Malthus's economic model, combining the concept of the diminishing average product with a theory of population growth, helps explain the long, flat part of history's hockey stick. It shows why (under the conditions Malthus described) technological improvement does not raise income permanently.

Concepts and models introduced and applied in Unit 1

- **Gross domestic product (GDP)**, growth in GDP per capita, global **income inequality**, and **climate change**
- **Technology**, **factors of production** and **production functions**; the **continuous technological revolution** (such as the **Industrial Revolution** in Britain)
- **Capitalism** is an **economic system** that is characterized by three **institutions**: **private property**, **markets**, and **firms**
- **Structural transformation**, whereby a part of the economy organized on capitalist lines expands and other sectors shrink
- The **Malthusian model**, diminishing **average product** of labour with **subsistence level** output being the **equilibrium**

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